An 88.8dB-SNDR 6-MS/s Pipelined SAR ADC with A Closed-Loop Dynamic Amplifier Featuring Highly-Linear Full-Scale Output Swing

Zongnan Wang[1], Haoyang Luo[1],
Rongjia Bao[1], Lu Jie[2], Xiyuan Tang[1]

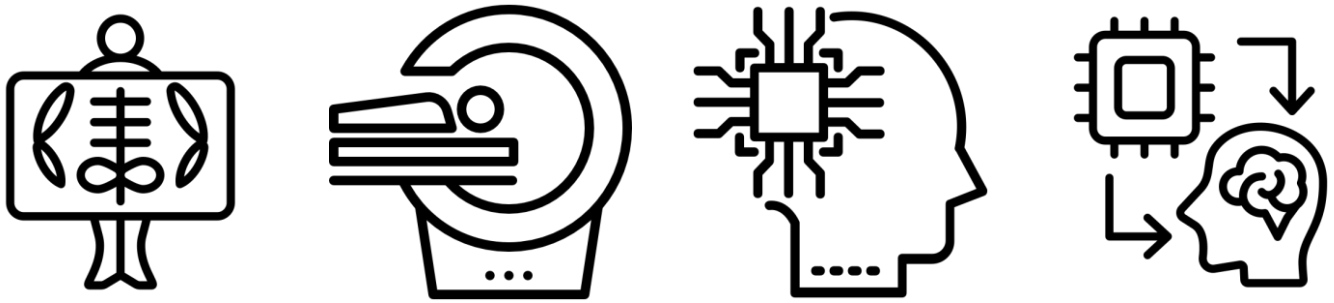
[1] Peking University, Beijing, China
[2] Tsinghua University, Beijing, China

Outline

- Motivation
- Proposed Closed-loop Dynamic Amplifier
  - Floating Gm-ratio-based Amplifier
  - Charge-pump-assisted Cascoded FIA
- Pipelined SAR ADC Circuit Implementation
  - System-level Chopping-assisted Auto-zero
- Measurement Results
- Conclusion

Motivation

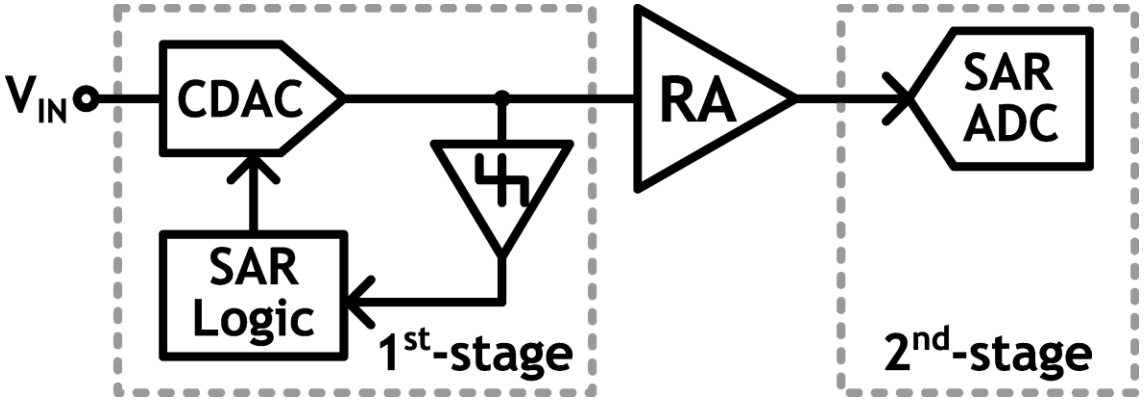
- High-resolution and high-speed ADC for time-multiplexing applications
  - Medical imaging
  - Neural probe
  - …



Motivation

- High-resolution and high-speed ADC for time-multiplexing applications
  - Medical imaging
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- Pipelined SAR ADC
  - Suppressed backend noise ➔ High resolution
  - Pipelined operation ➔ High speed

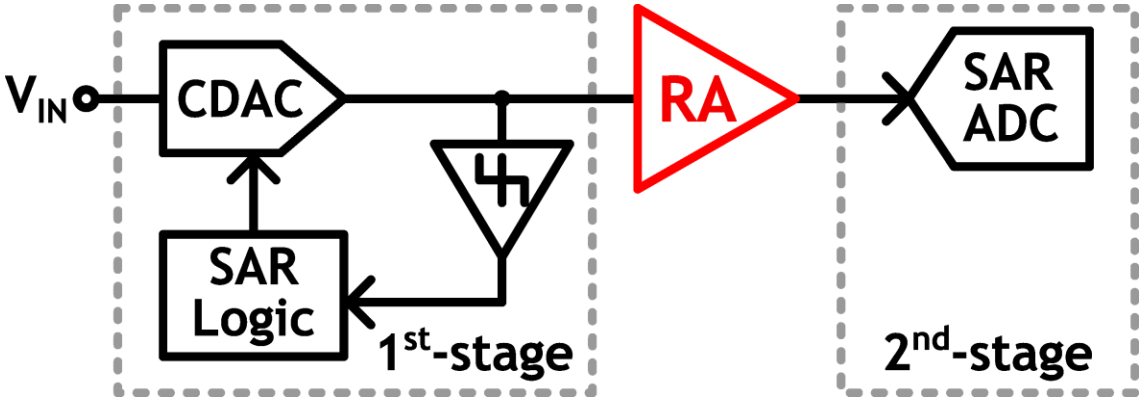


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- High-resolution and high-speed ADC for time-multiplexing applications
  - Medical imaging
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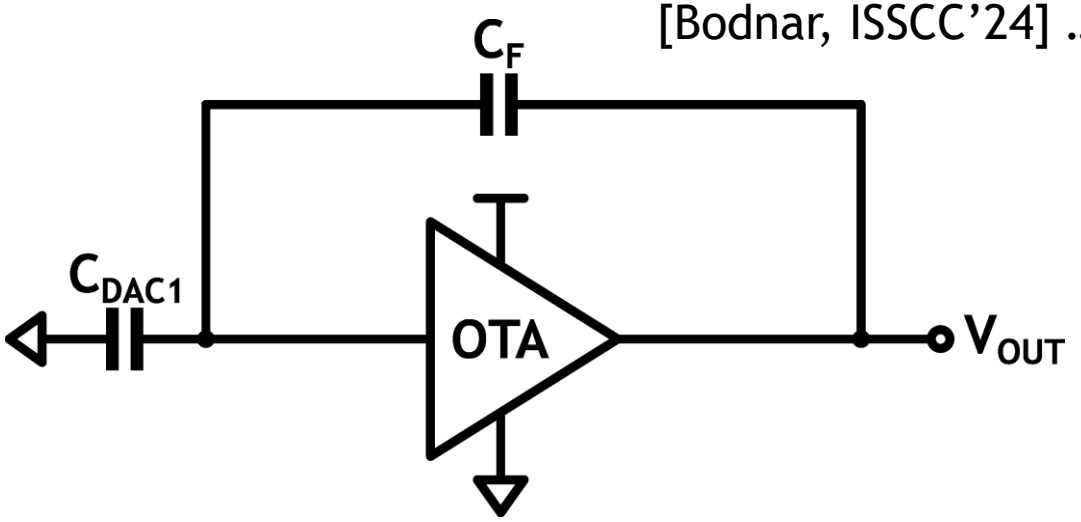
The key to high performance lies in the **RA**



OTA-based RA

- OTA-based RA

[Bannon, VLSI'14]
[Hummerston, VLSI'17]
[Bodnar, ISSCC'24] …

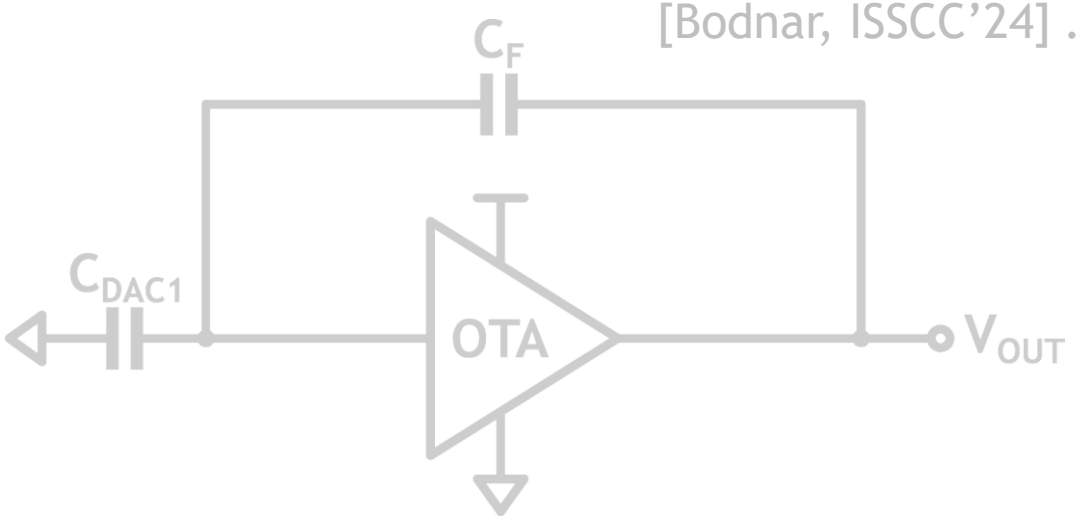


  - Robust closed-loop performance ☺
  - Static power consumption ☹
  - Scaling unfriendly ☹

Ring Amplifier

- OTA-based RA

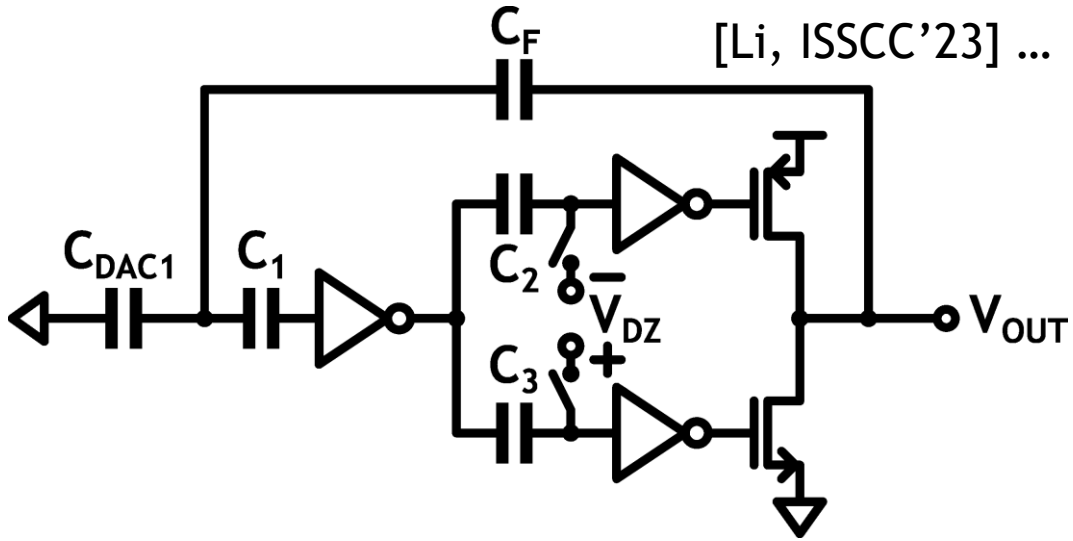
[Bannon, VLSI'14]
[Hummerston, VLSI'17]
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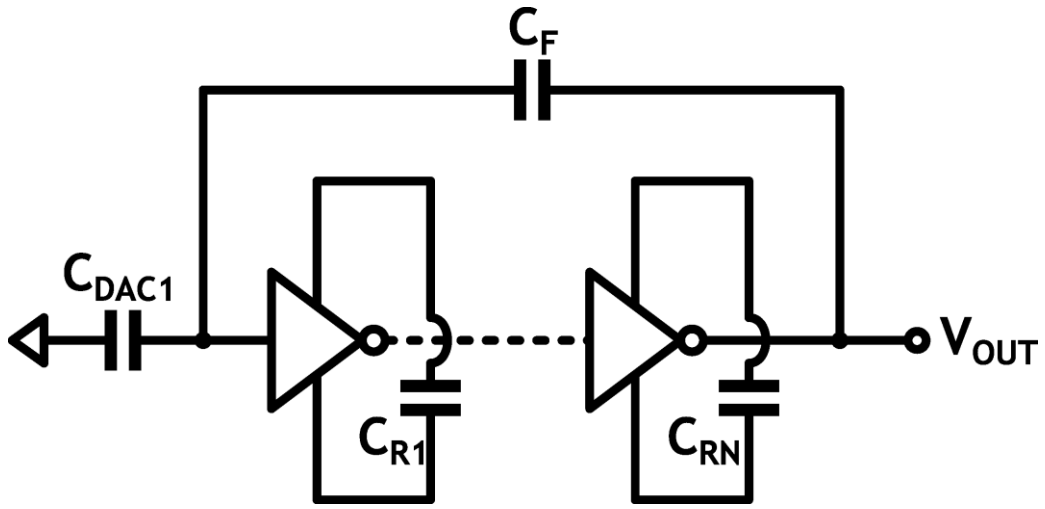
[Hershberg, JSSC'12]
[Lim, VLSI'17]
[Zhan, ISSCC'22]
[Li, ISSCC'23] …



- High speed and slew efficiency ☺
- PVT-sensitive stability ☹
- Static 1st-stage amp. ☹

FIA-based RA

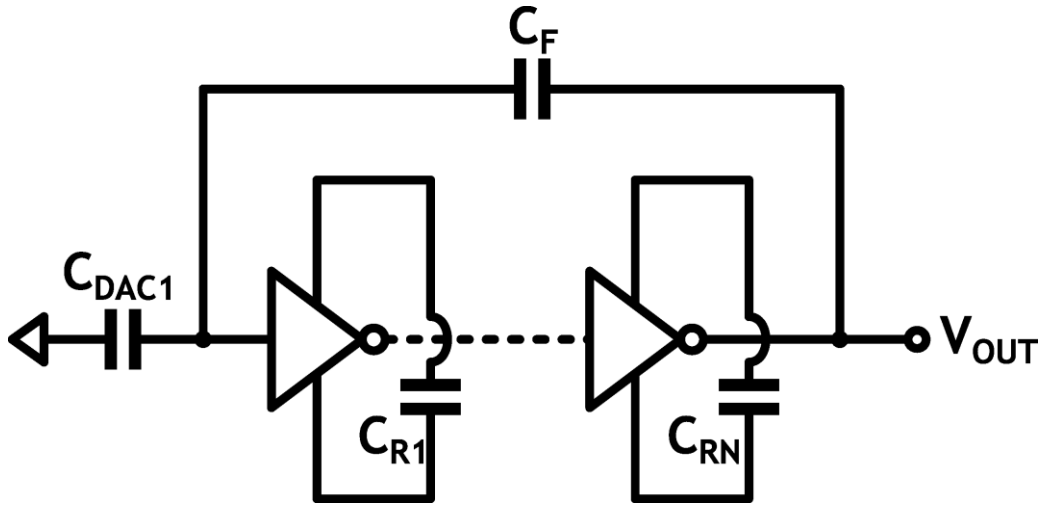
- FIA-based RA [Tang, ISSCC'21] [Song, VLSI'23] …



  - High energy efficiency ☺
  - PVT-robust stability ☺
  - Limited speed and output swing ☹

FIA-based RA

- FIA-based RA [Tang, ISSCC'21] [Song, VLSI'23] …



How to improve the speed and output swing ?

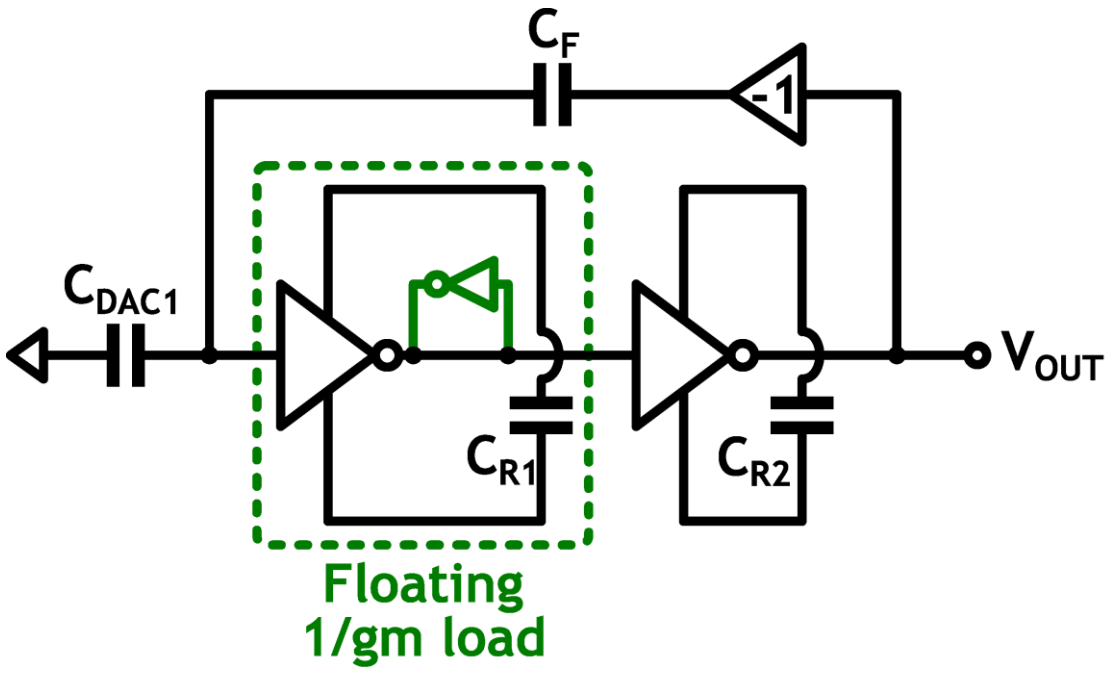
- High energy efficiency ☺
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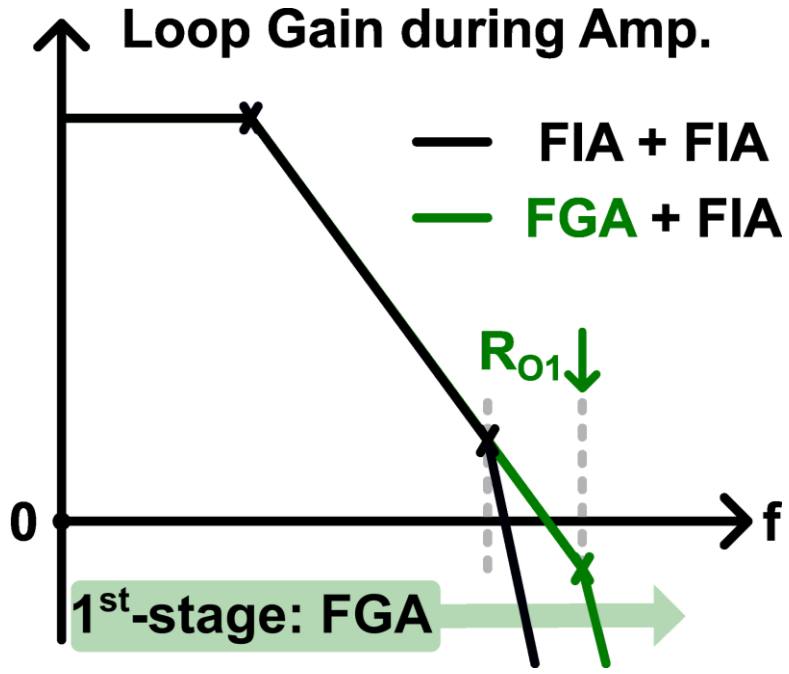
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Floating Gm-ratio-based Amplifier

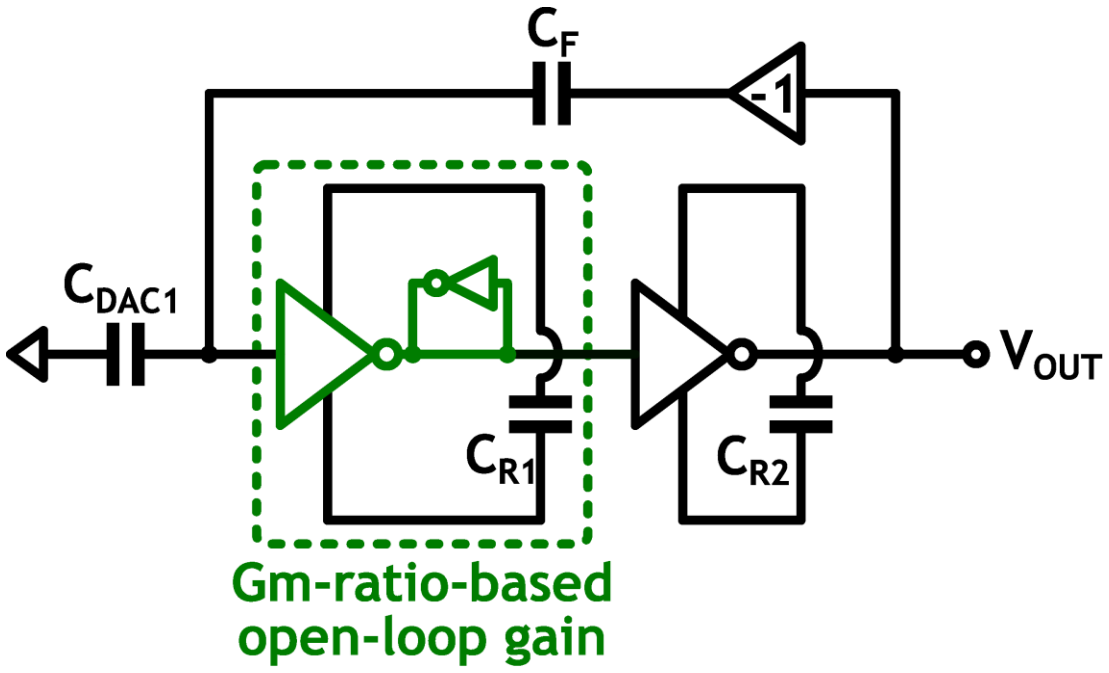
- 1st-stage Floating Gm-ratio-based Amplifier (FGA)
  - High-frequency non-dominant pole ➔ Improved speed and stability ☺

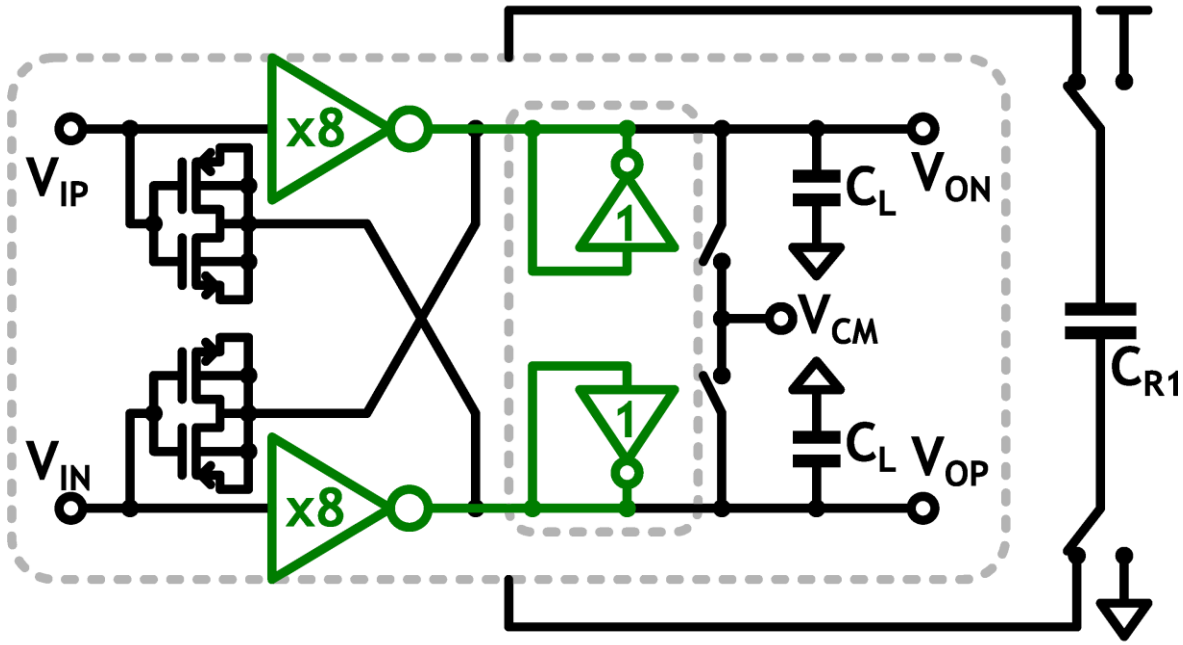




Floating Gm-ratio-based Amplifier

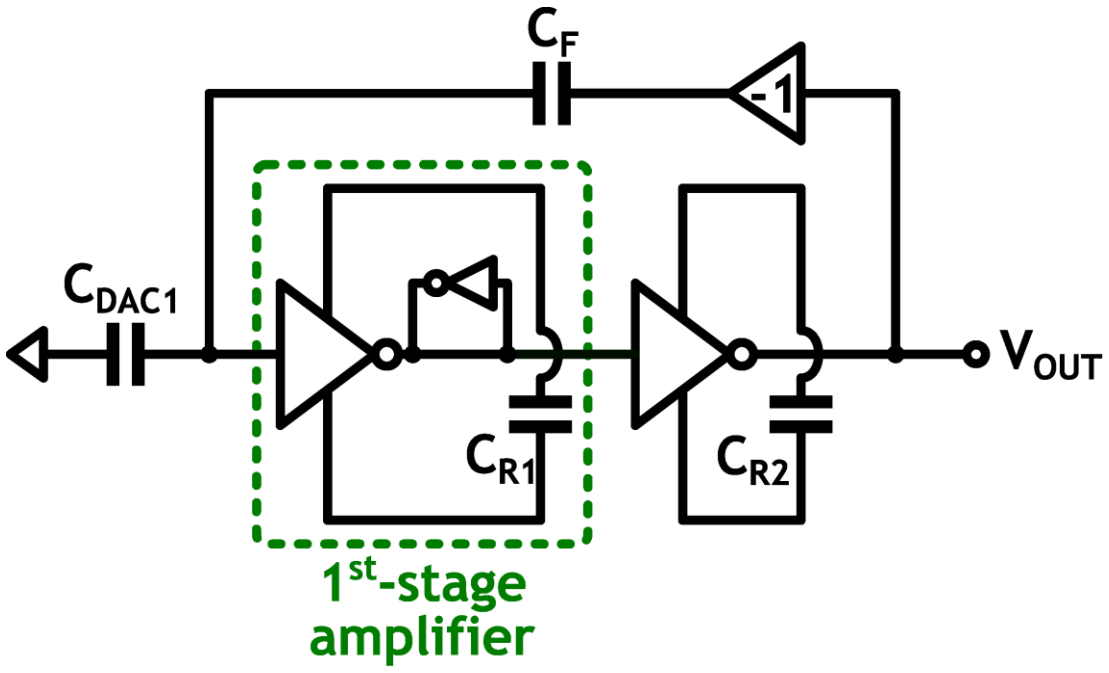
- 1st-stage Floating Gm-ratio-based Amplifier (FGA)
  - High-frequency non-dominant pole ➔ Improved speed and stability ☺
  - Gm-ratio-based open-loop gain ➔ Improved robustness ☺

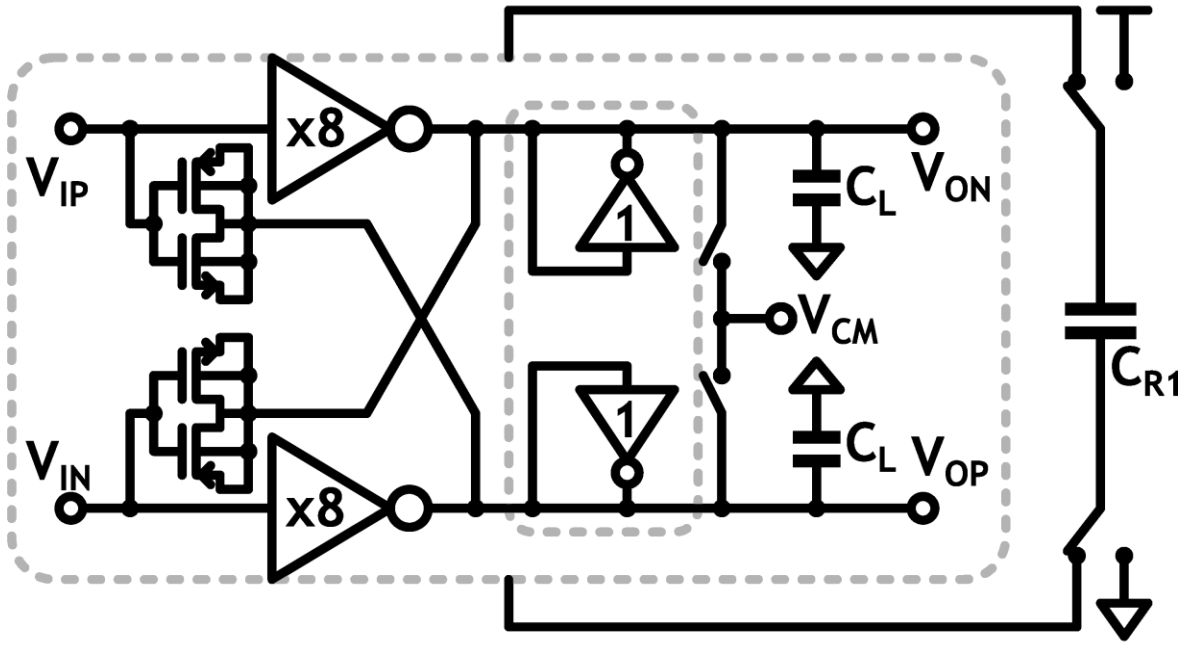




Floating Gm-ratio-based Amplifier

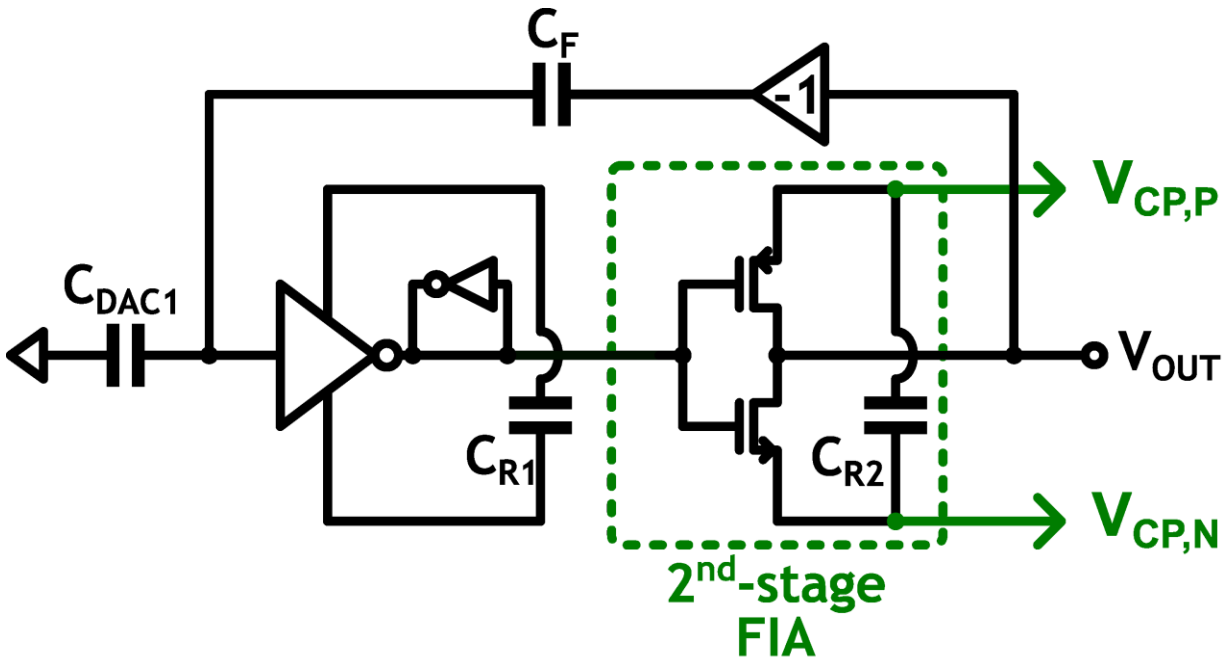
- 1st-stage Floating Gm-ratio-based Amplifier (FGA)
  - High-frequency non-dominant pole ➔ Improved speed and stability ☺
  - Gm-ratio-based open-loop gain ➔ Improved robustness ☺
  - Small output swing ➔ Relaxed linearity requirement ☺

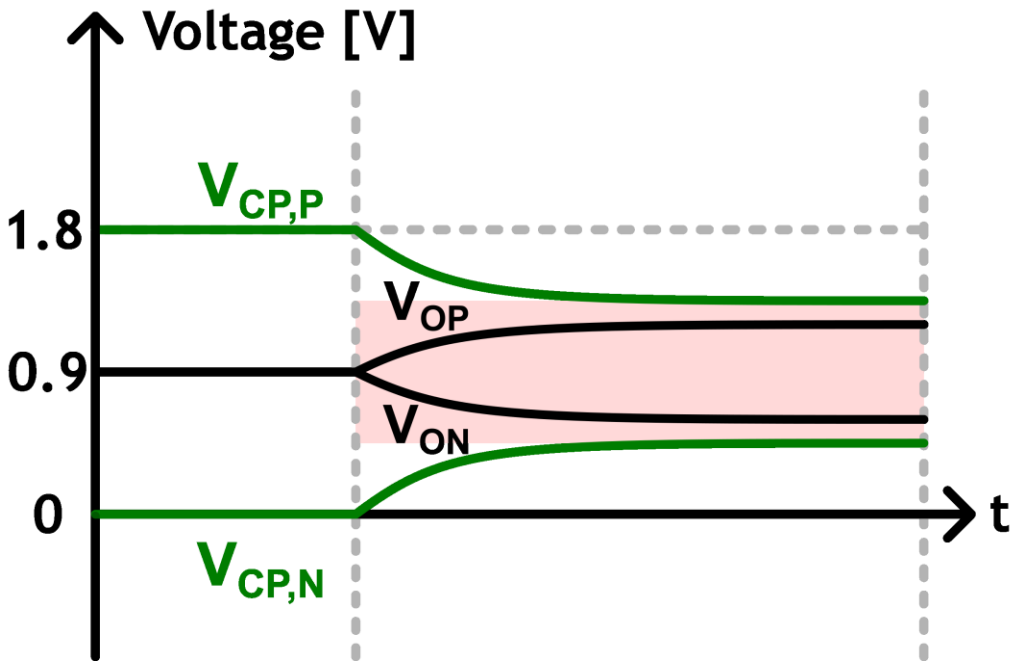




Output-stage amplifier

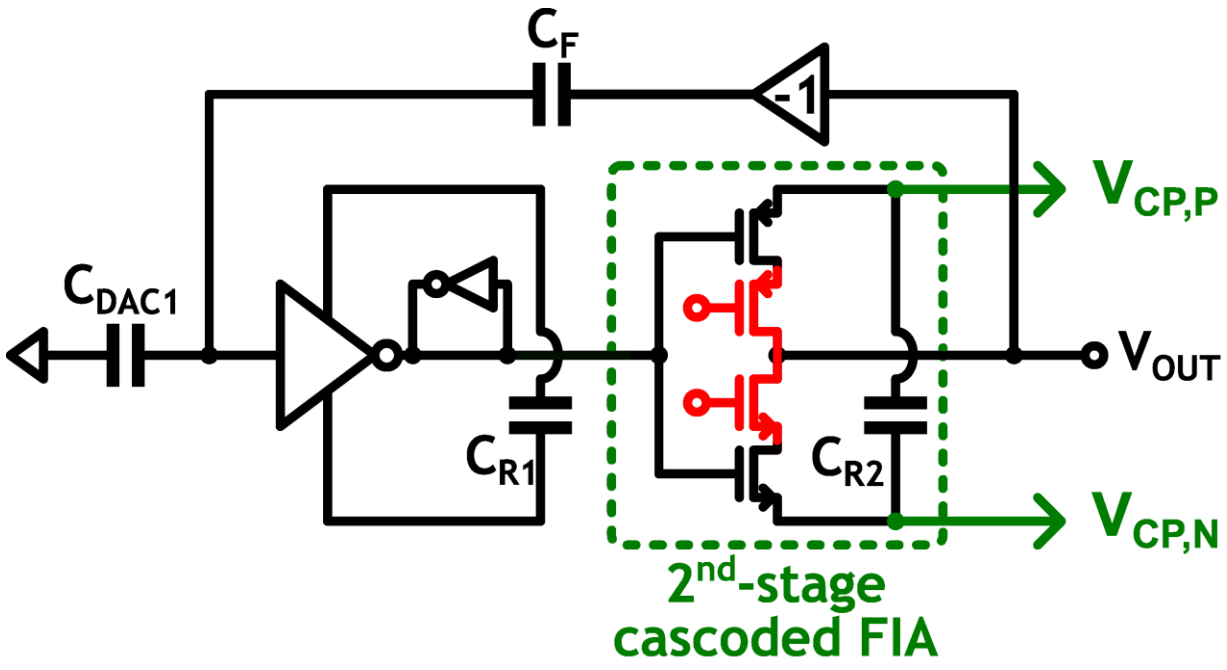
- 2nd-stage FIA
  - Reducing supply ➔ Limited output swing and slew rate ☹

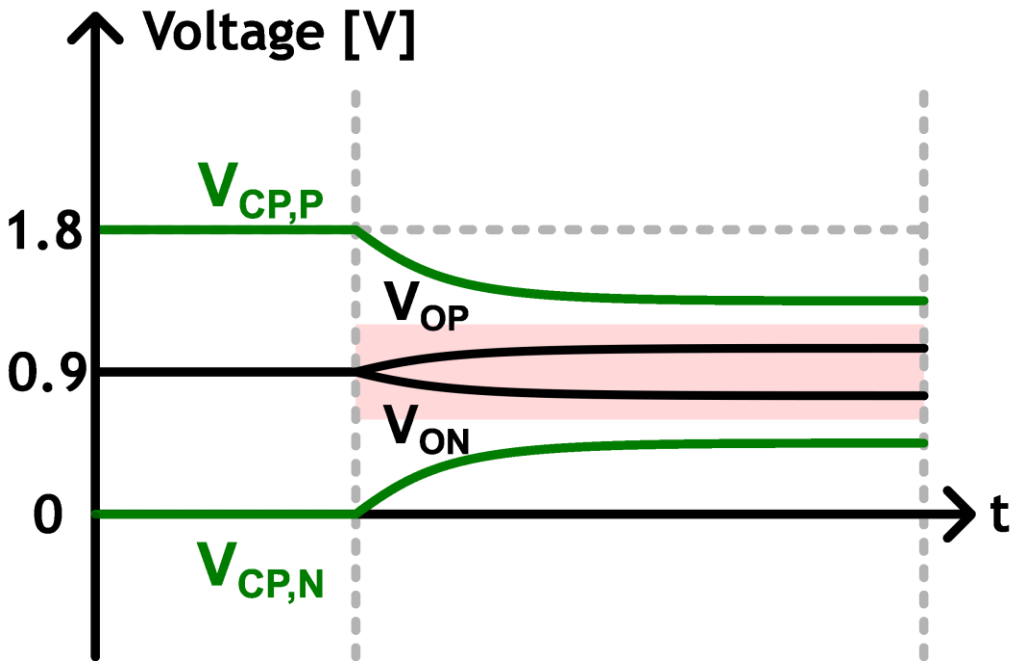




Output-stage amplifier

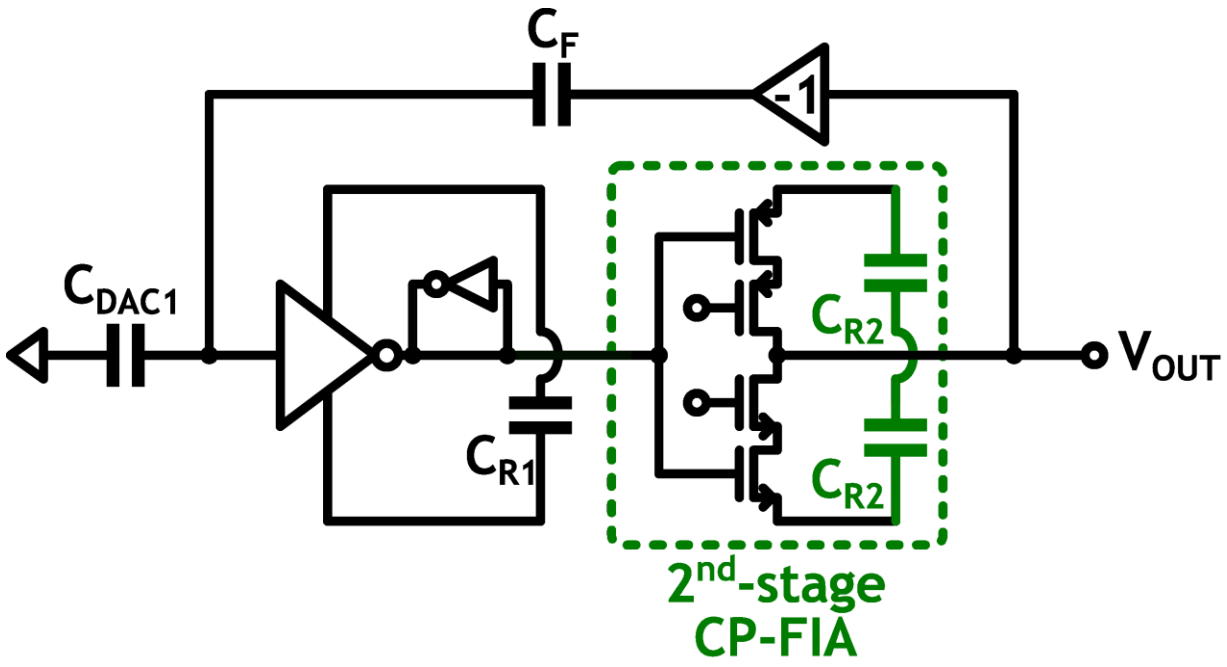
- 2nd-stage Cascoded FIA
  - Reducing supply ➔ Limited output swing and slew rate ☹
  - Cascoded output stage ➔ Improved loop gain and stability ☺
    ➔ Worsened output swing and slew rate ☹

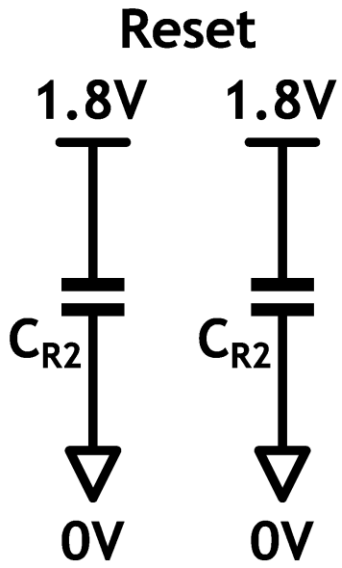




Charge-pump-assisted Cascoded FIA

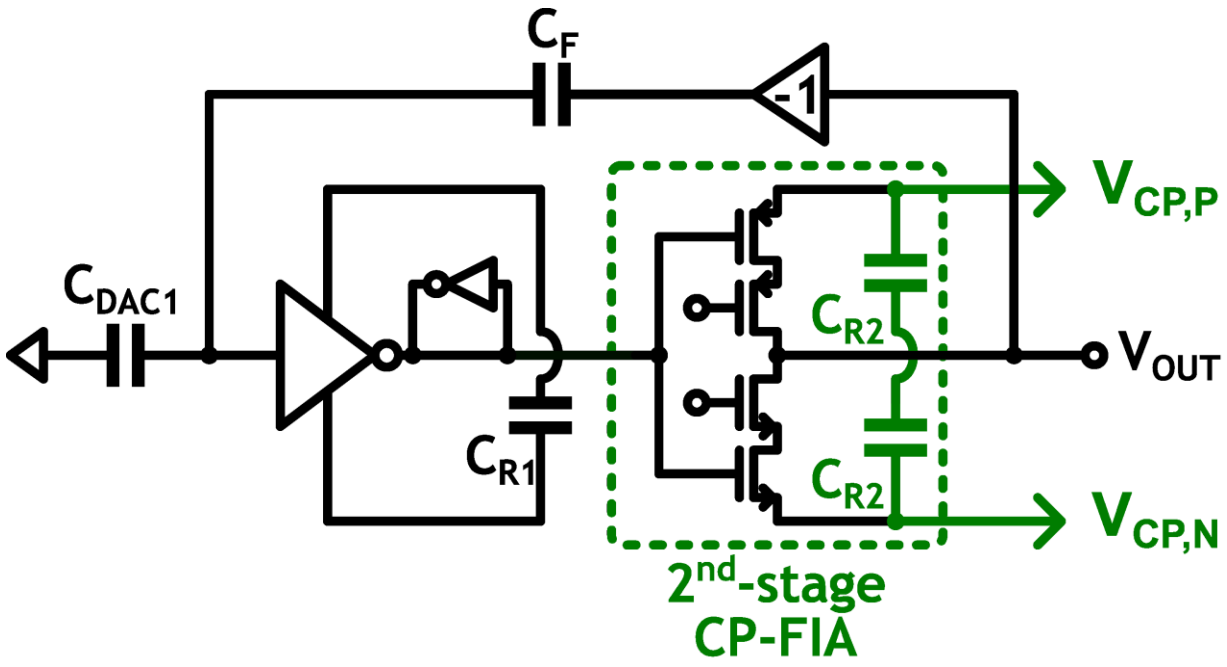
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Reset phase: charge two reservoir capacitors with core supply voltage (1.8V)

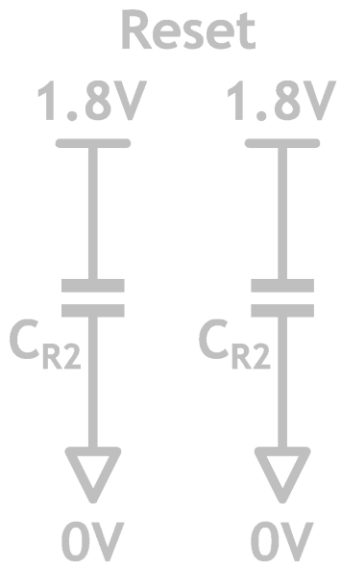


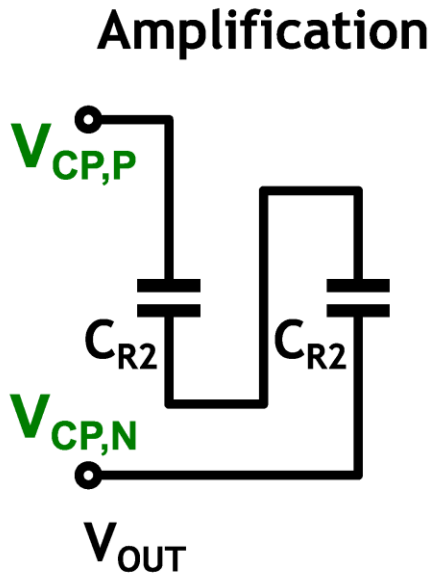


Charge-pump-assisted Cascoded FIA

- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Amplification phase: cascade two reservoir capacitors for doubled supply

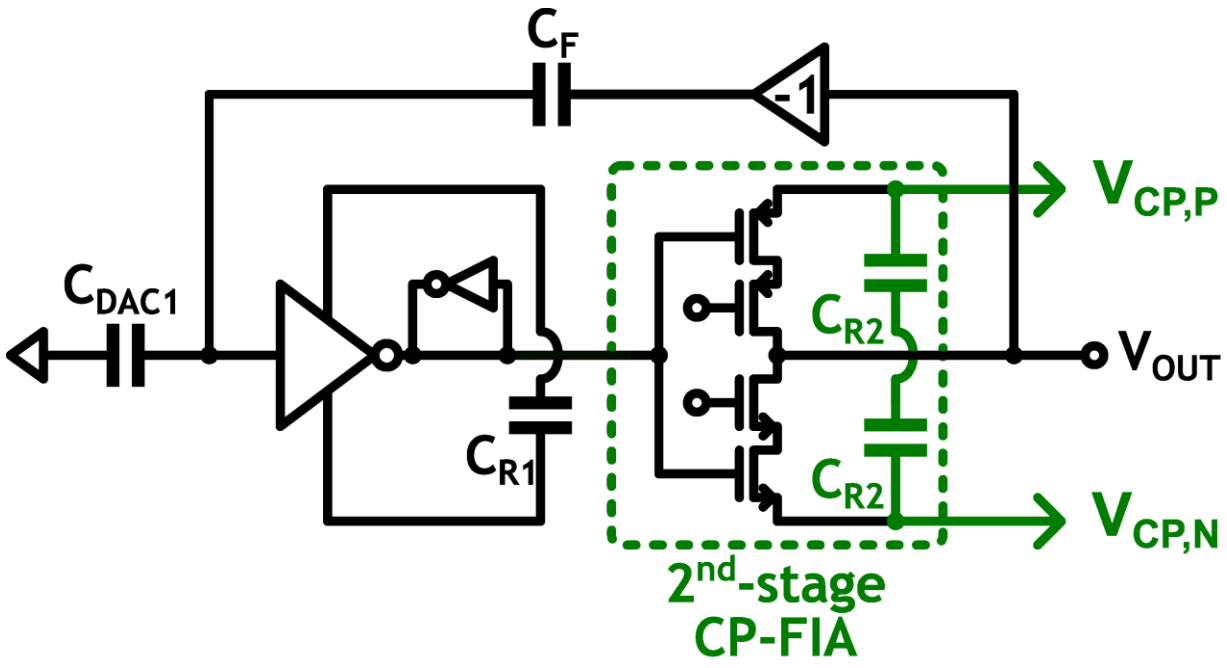


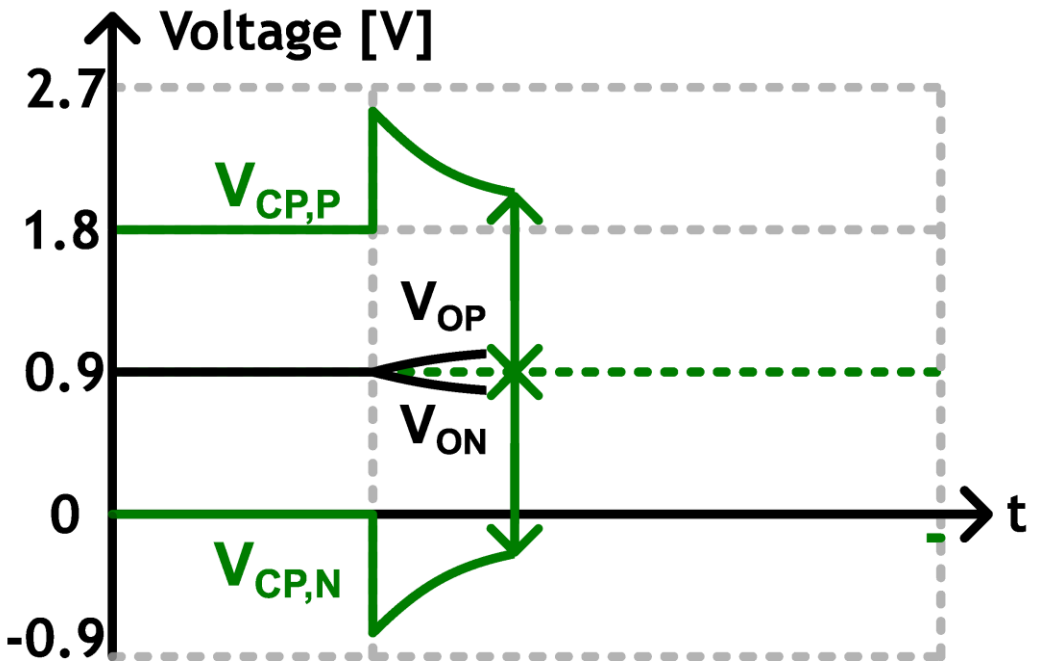




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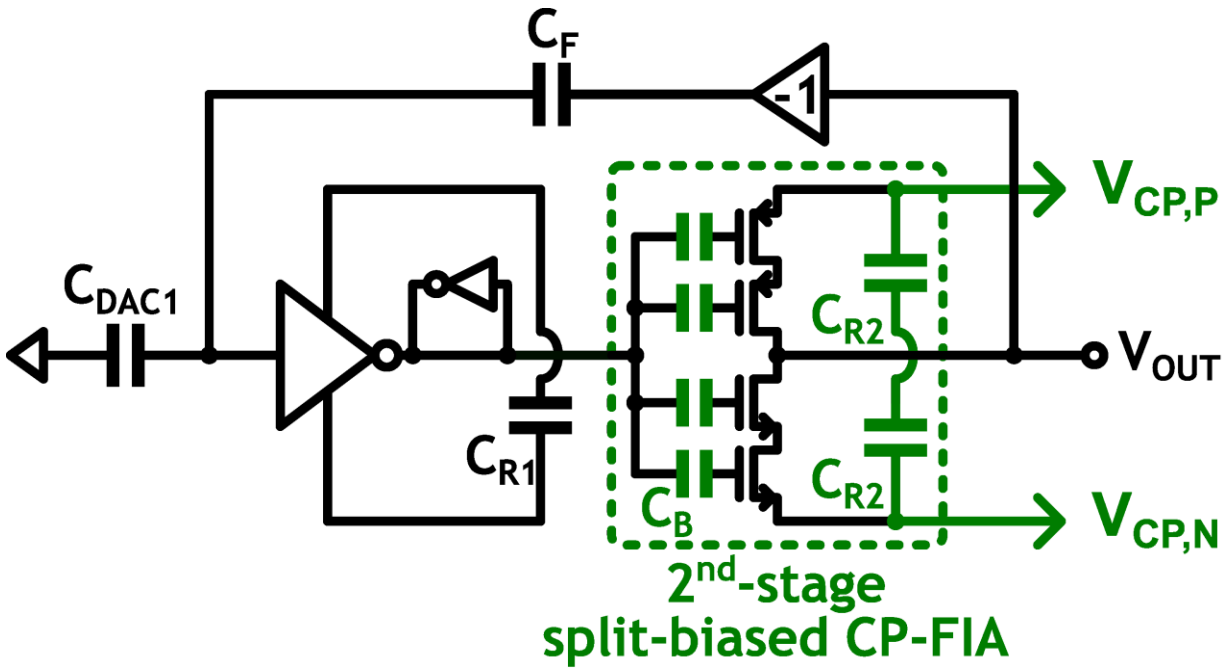
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Amplification phase: cascade two reservoir capacitors for doubled supply
  - Floating domain: center the doubled supply around V_{CM} (0.9V)

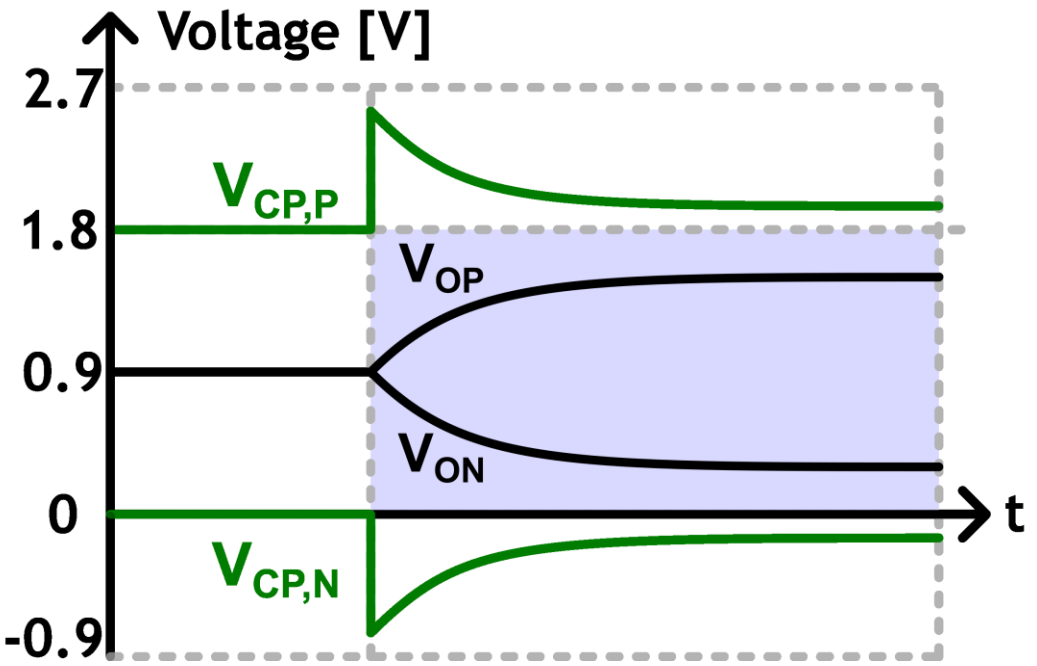




Charge-pump-assisted Cascoded FIA

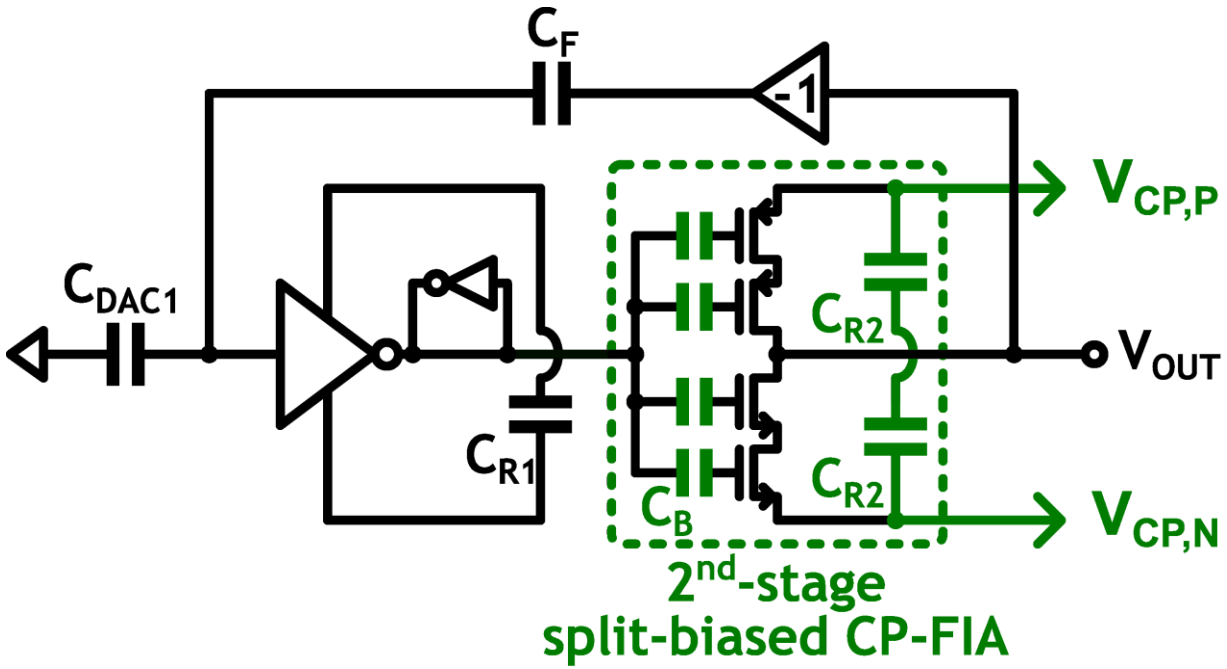
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
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  - Split-biased cascoded input pair: from saturation region to subthreshold region

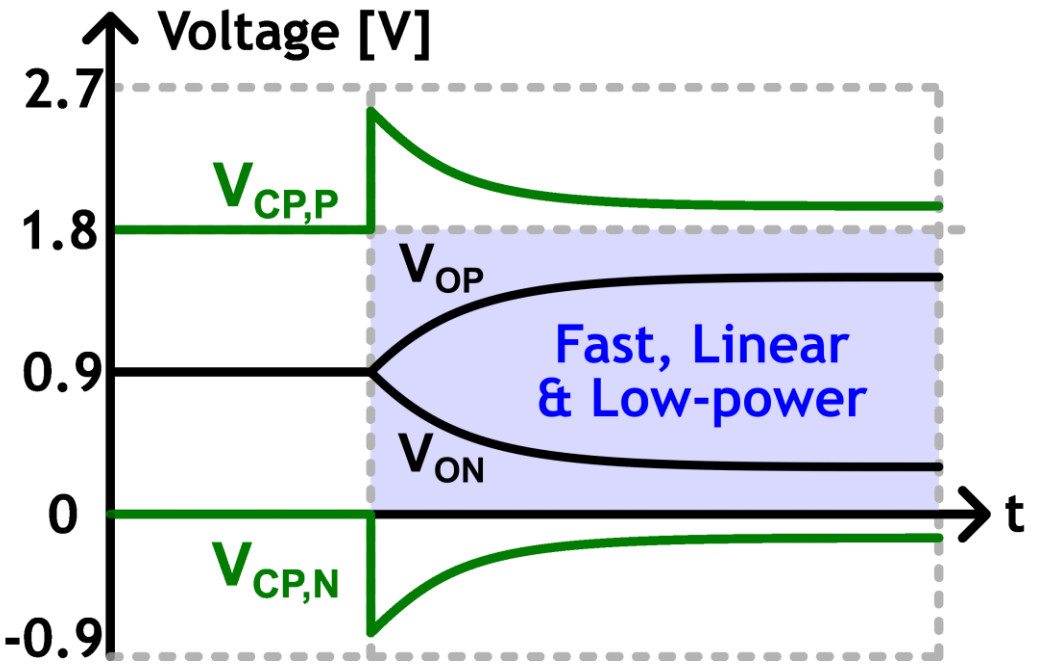




Charge-pump-assisted Cascoded FIA

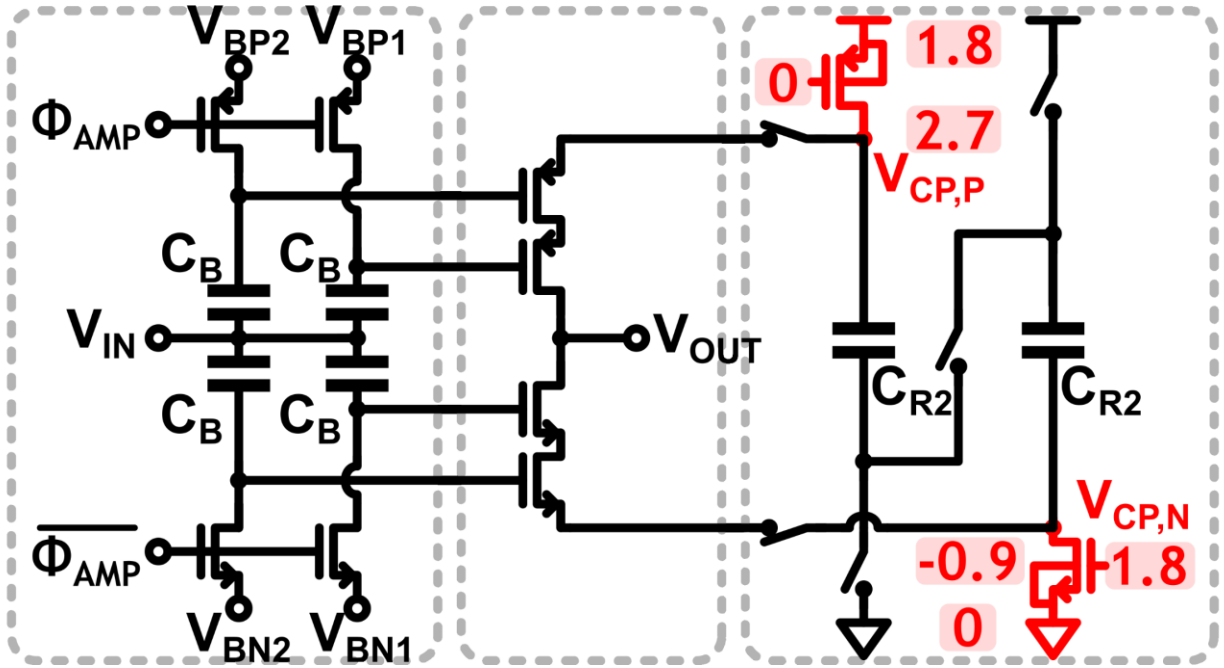
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Doubled supply ➔ Improved output swing and slew rate ☺
  - Floating domain ➔ Stable output CM voltage ☺
  - Self quenching ➔ Decaying bandwidth for low noise and power ☺

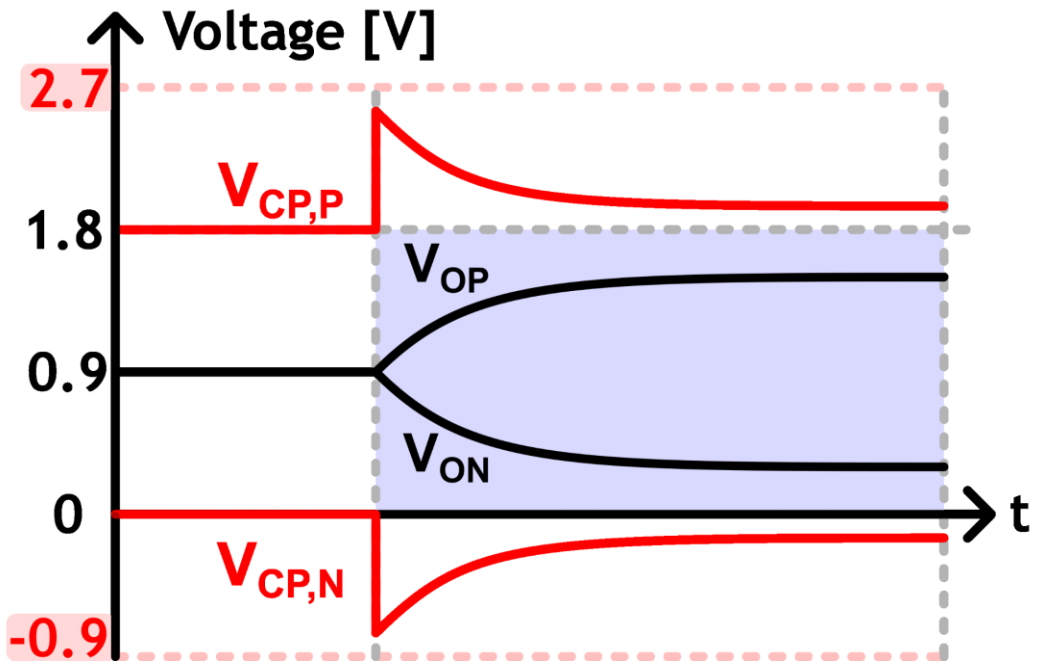




Charge-pump-assisted Cascoded FIA

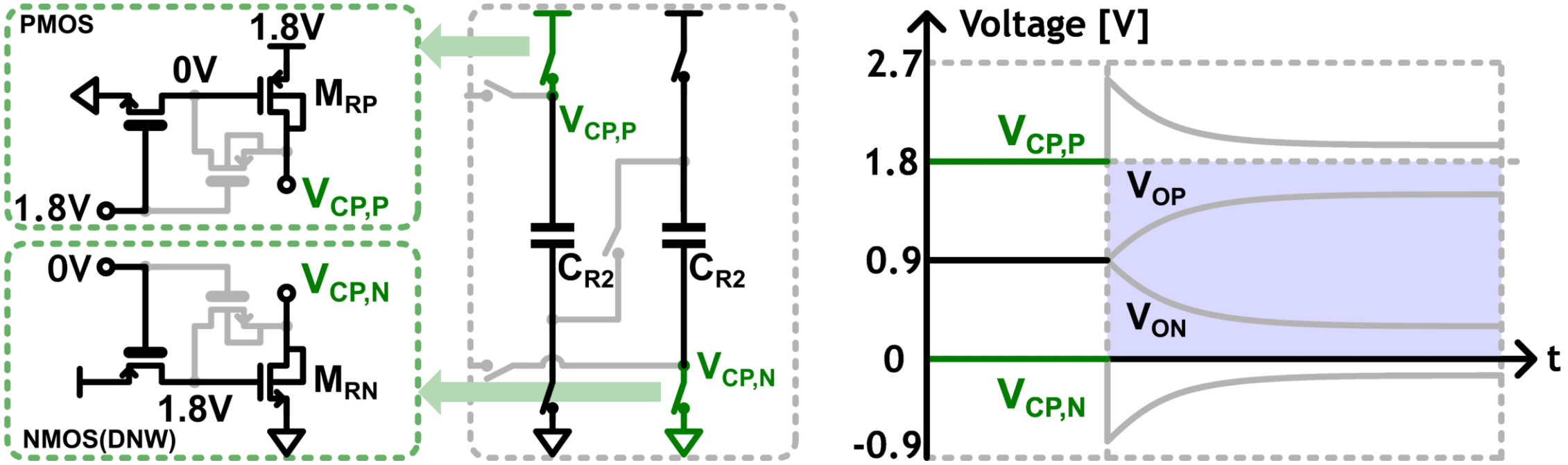
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Channel and junction leakage problems for switches with 2.7V/-0.9V ☹





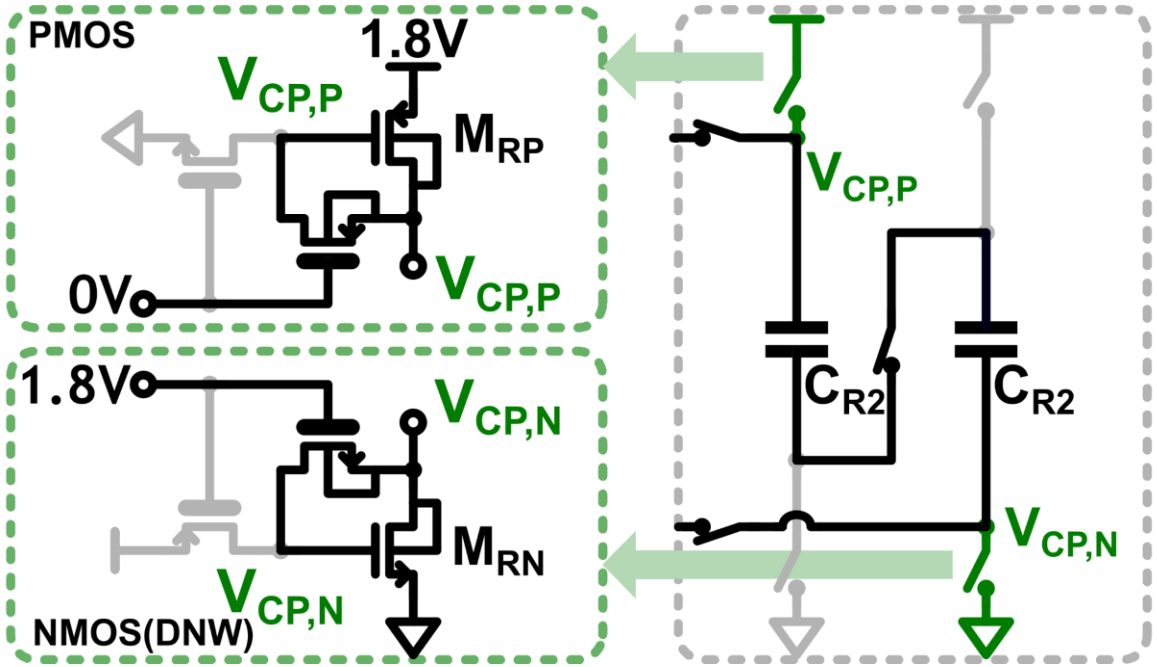
Charge-pump-assisted Cascoded FIA

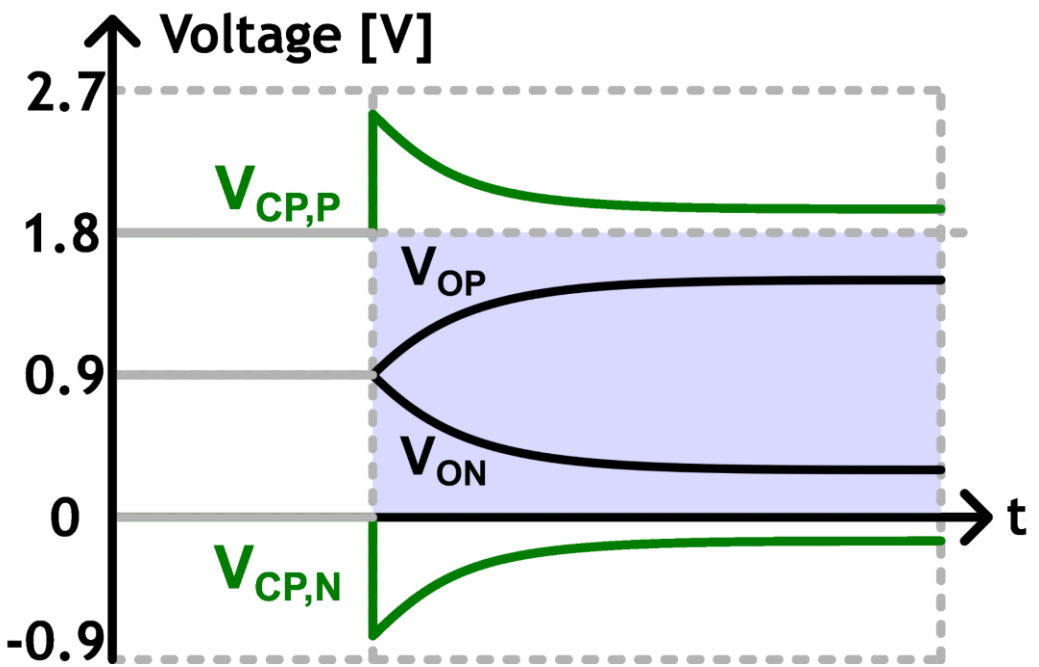
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Reset phase: connect the gate of PMOS/NMOS switches to 0V/1.8V



Charge-pump-assisted Cascoded FIA

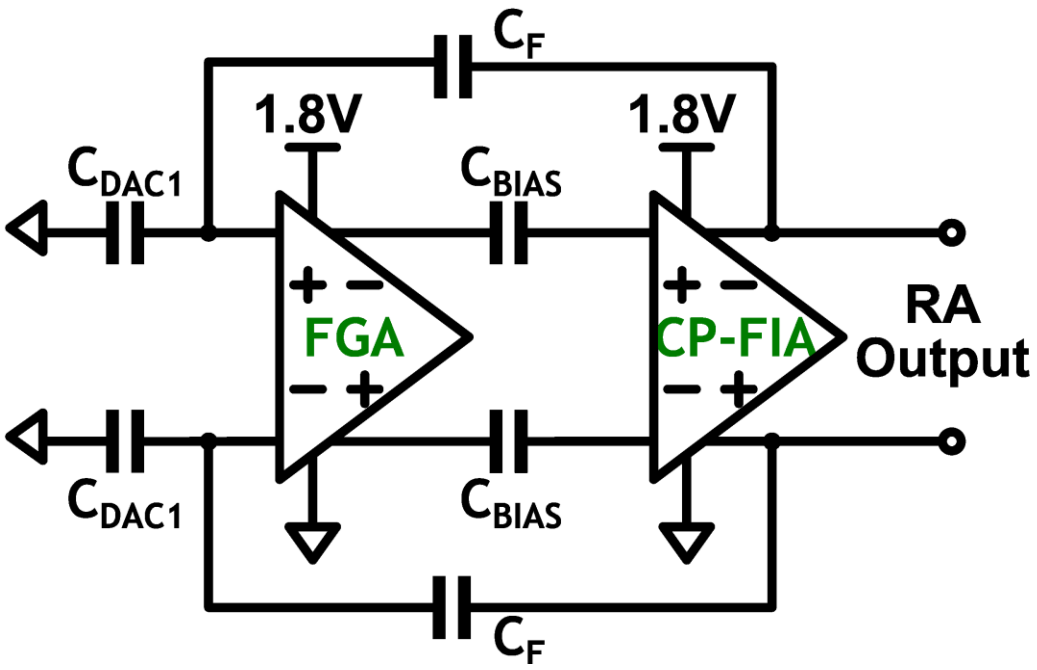
- 2nd-stage Charge-pump-assisted Cascoded FIA (CP-FIA)
  - Amplification phase: short both the gate and bulk of switches to the drain
  - No channel or junction leakage ☺

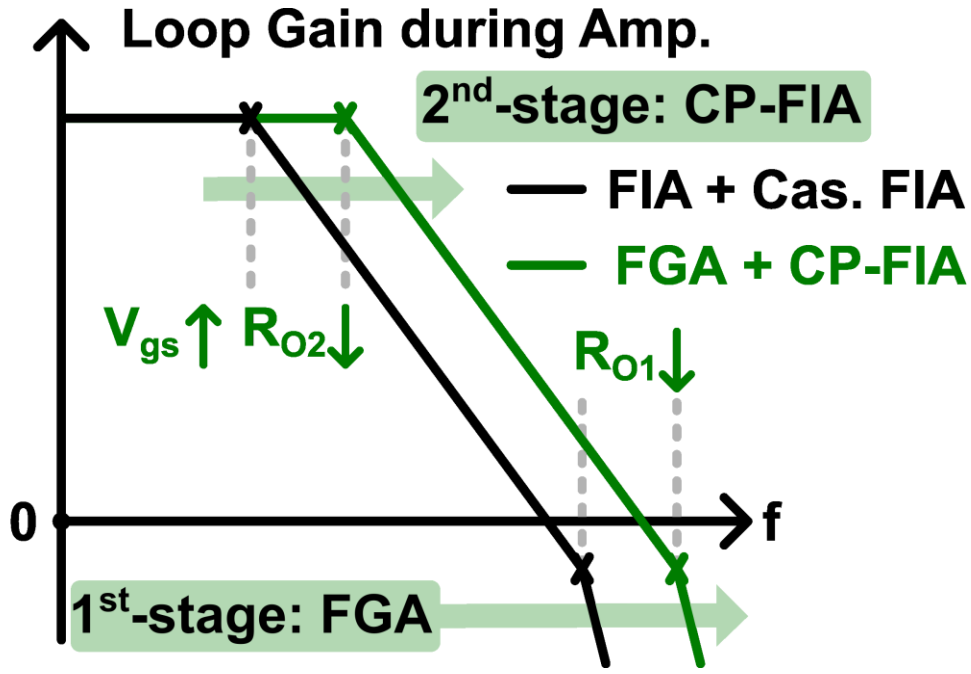




Closed-loop Dynamic Amplifier

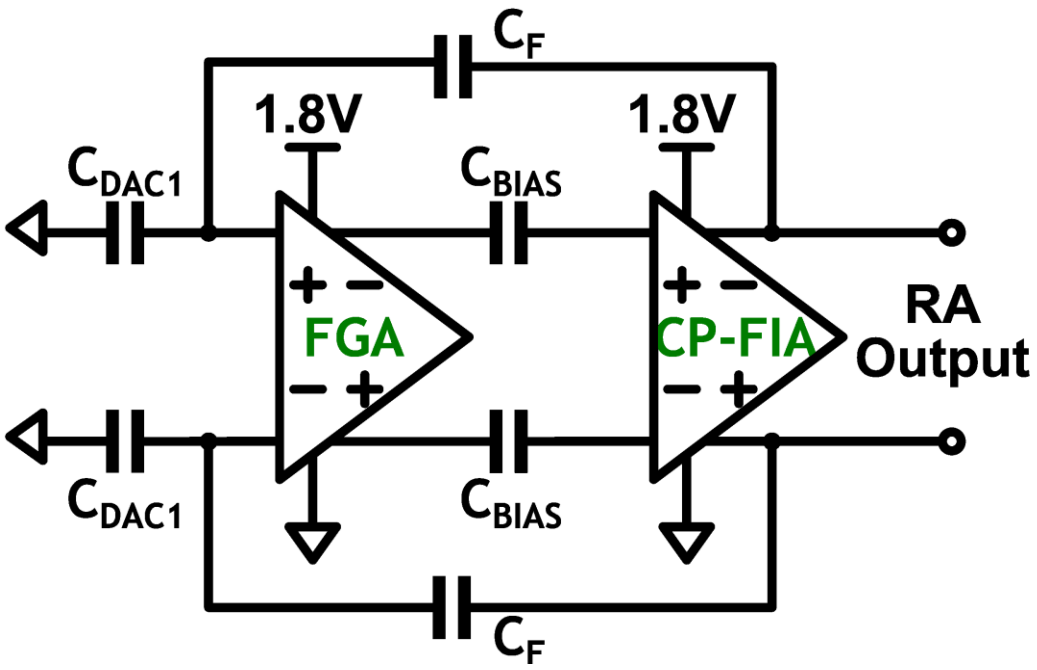
- 1st-stage FGA + 2nd-stage CP-FIA
  - High speed and robust stability ☺
  - High energy efficiency ☺

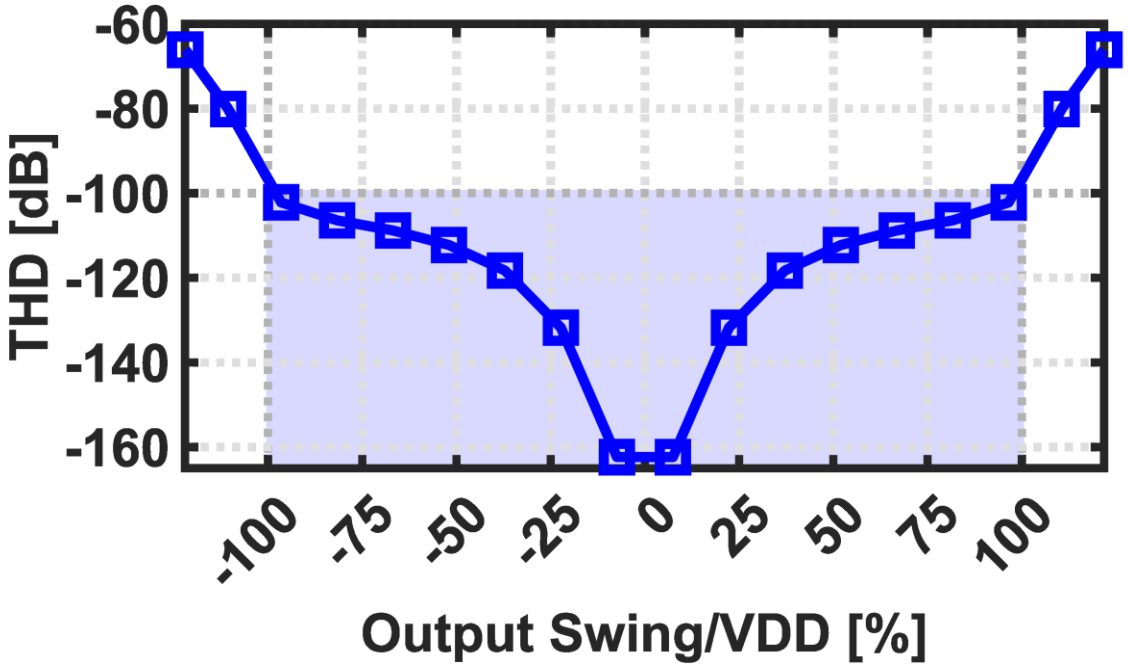




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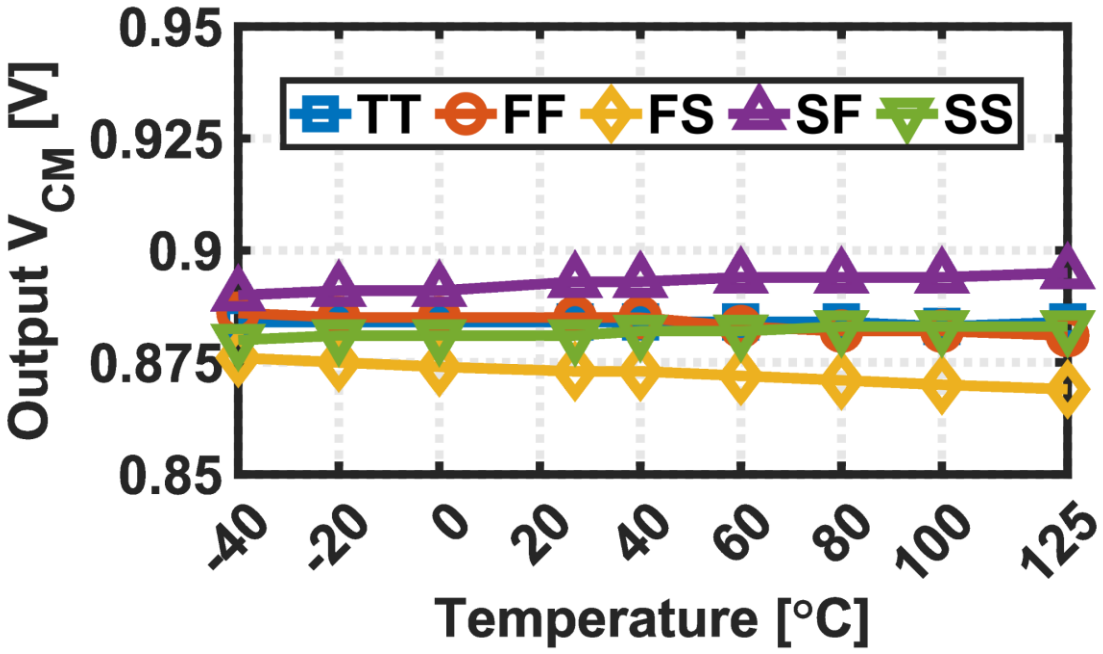
- 1st-stage FGA + 2nd-stage CP-FIA
  - <-100dB THD with a full-scale output swing ➔ Good linearity ☺

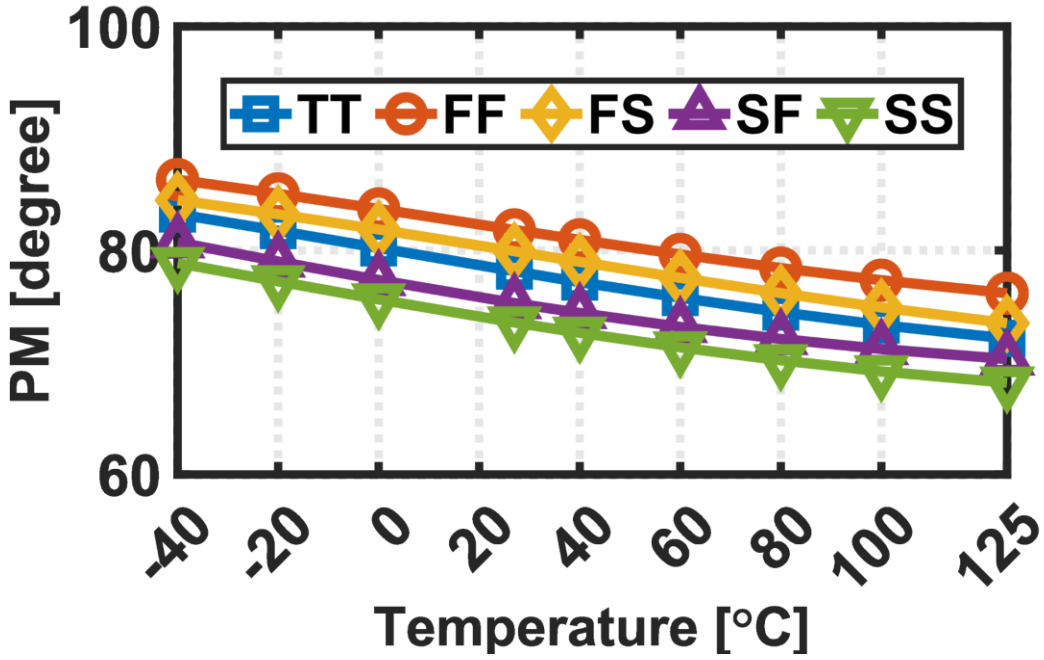




Closed-loop Dynamic Amplifier

- 1st-stage FGA + 2nd-stage CP-FIA
  - Stable output CM & >60 degrees PM over temperature and corner variations ☺



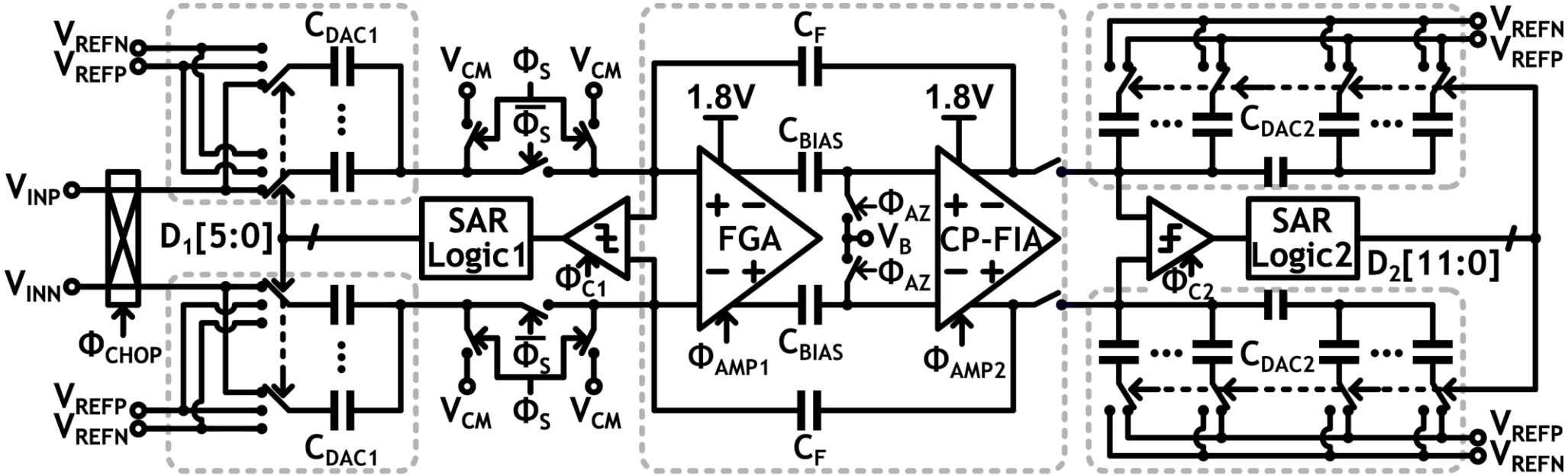


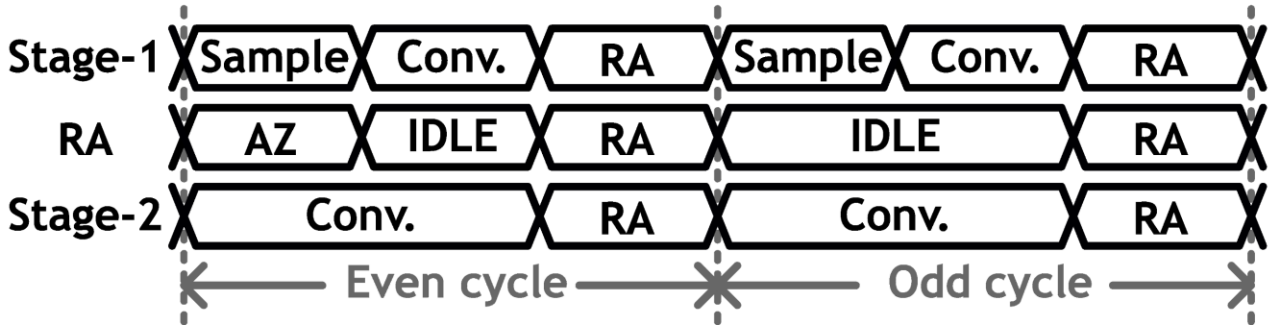
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Circuit Implementation

- Simplified Schematic

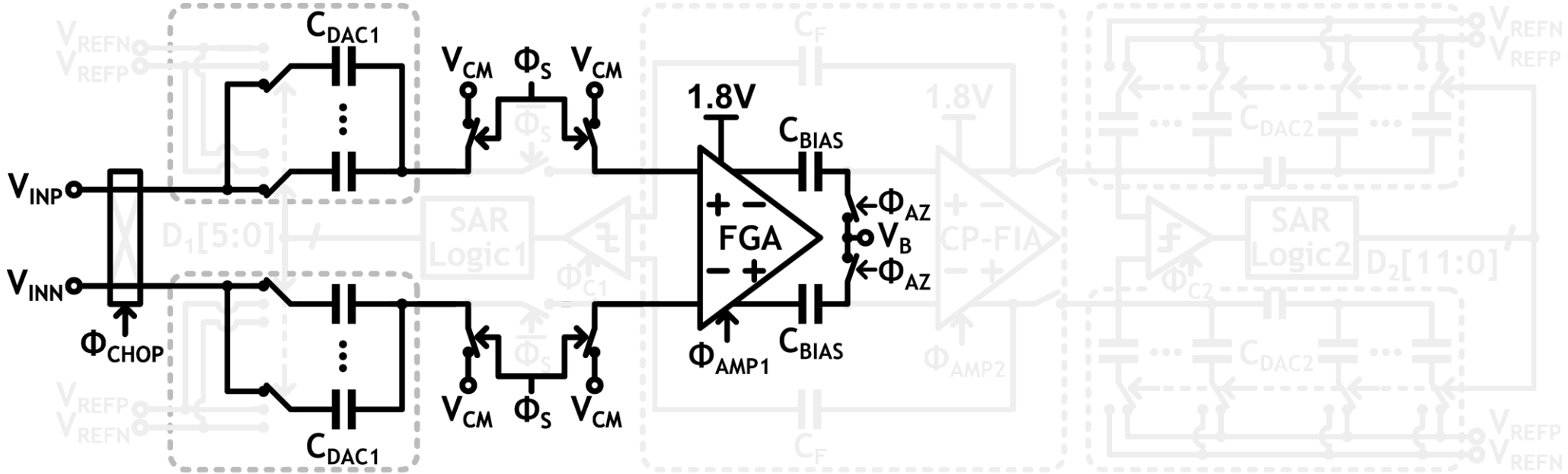


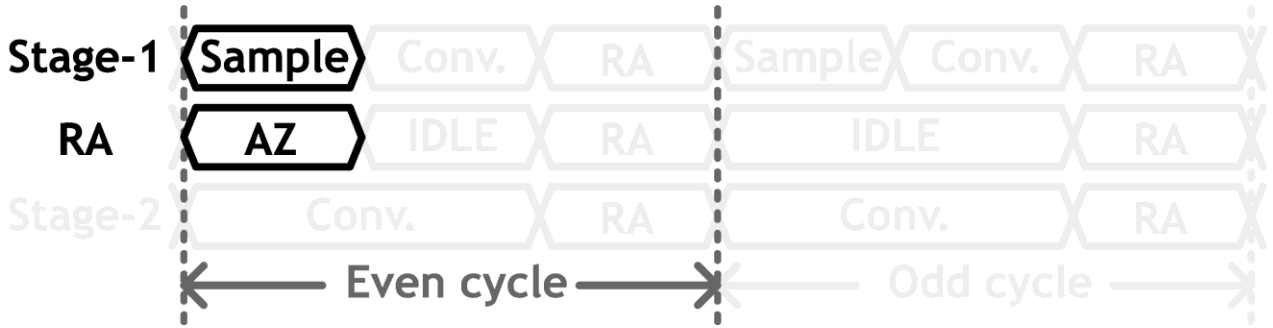


- 6b 1st-stage SAR
- 12b 2nd-stage SAR
- 32x inter-stage gain

Pipelined SAR ADC Operations

- Sample and AZ

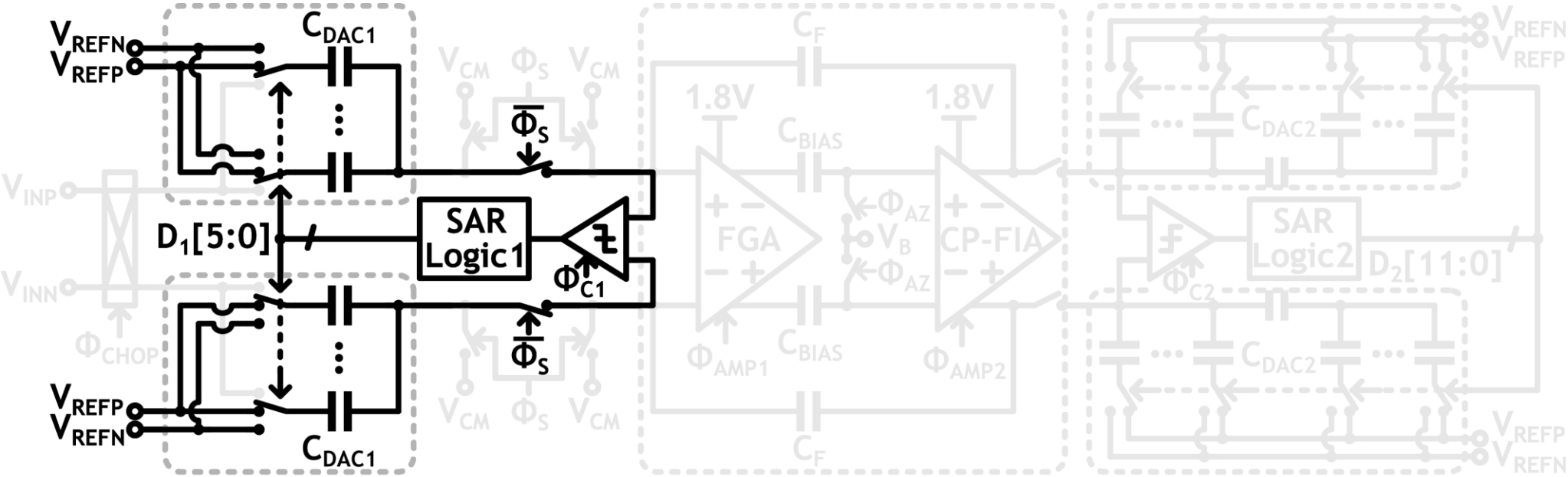


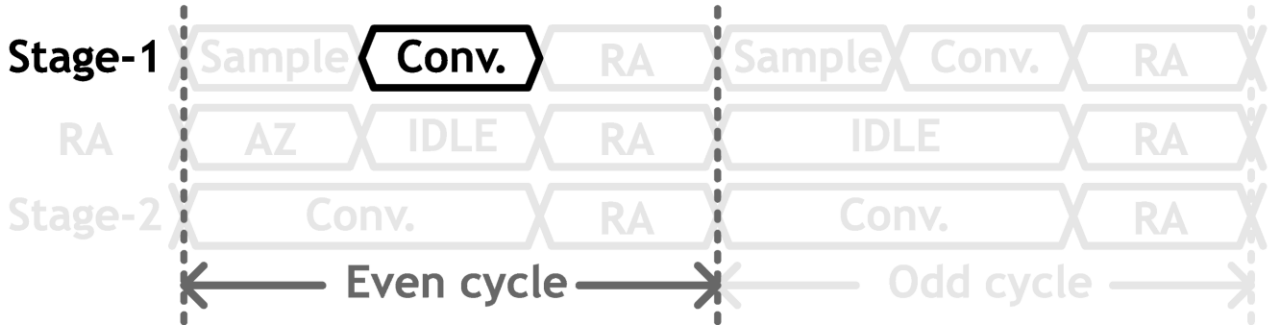


- Output AZ for 1st-stage FGA
- The right of C_{BIAS} is reset to V_B to split-bias CP-FIA

Pipelined SAR ADC Operations

- 1st-stage Conversion

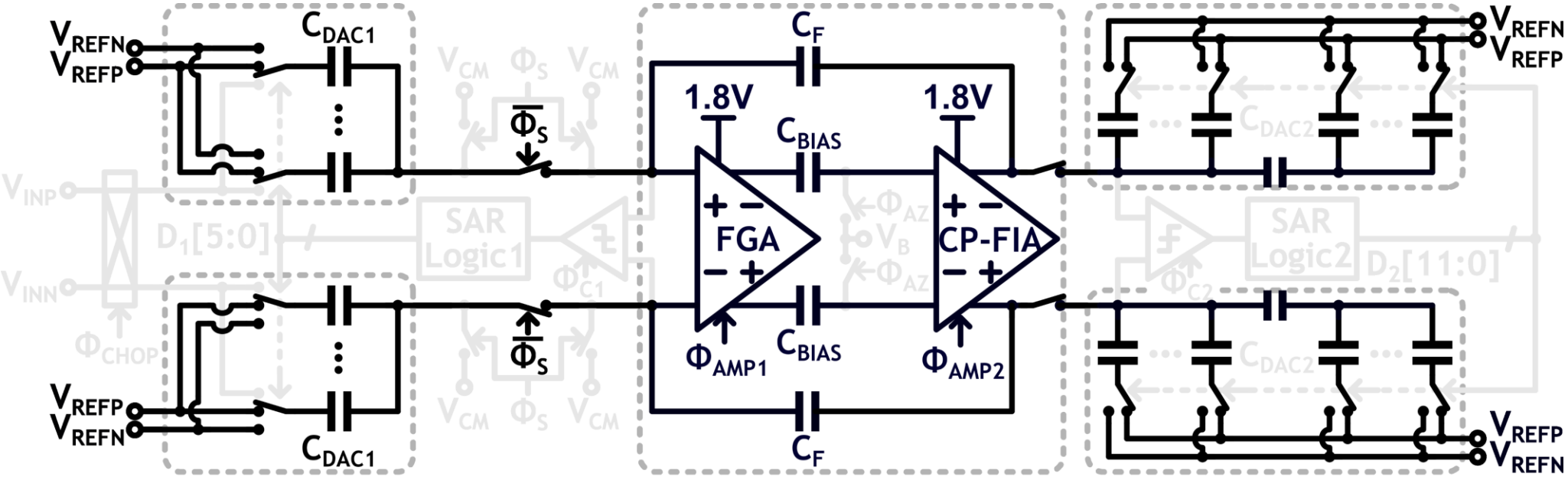


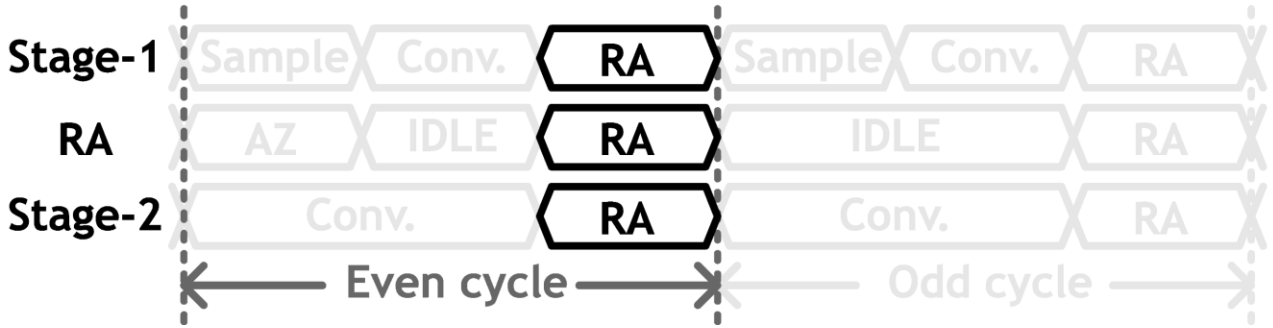


- 6b SAR conversion (Async)

Pipelined SAR ADC Operations

- Residue Amplification

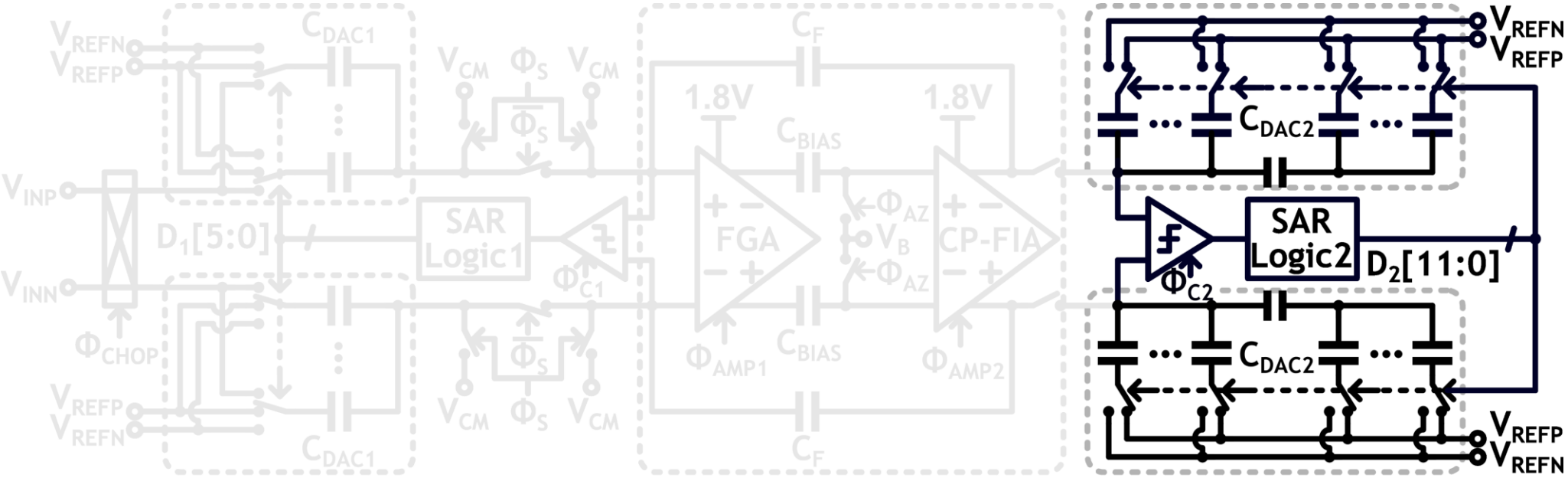


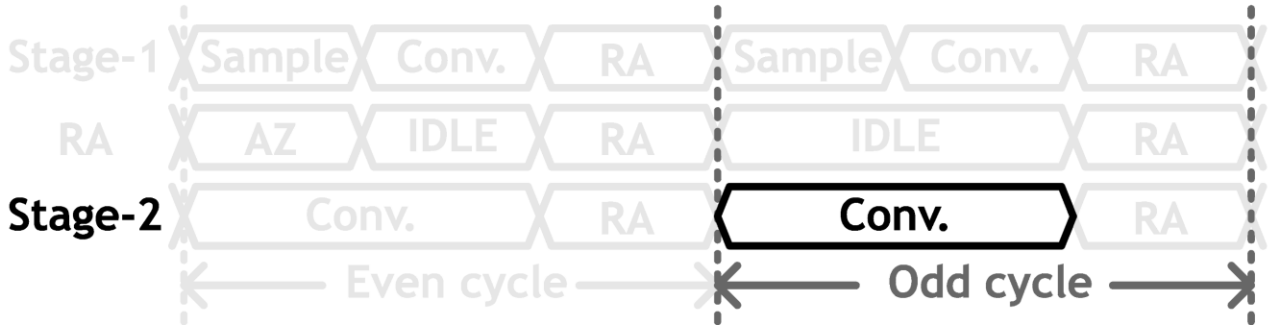


- 1st-stage FGA + 2nd-stage CP-FIA with embedded C_{BIAS} for split biasing

Pipelined SAR ADC Operations

- 2nd-stage Conversion

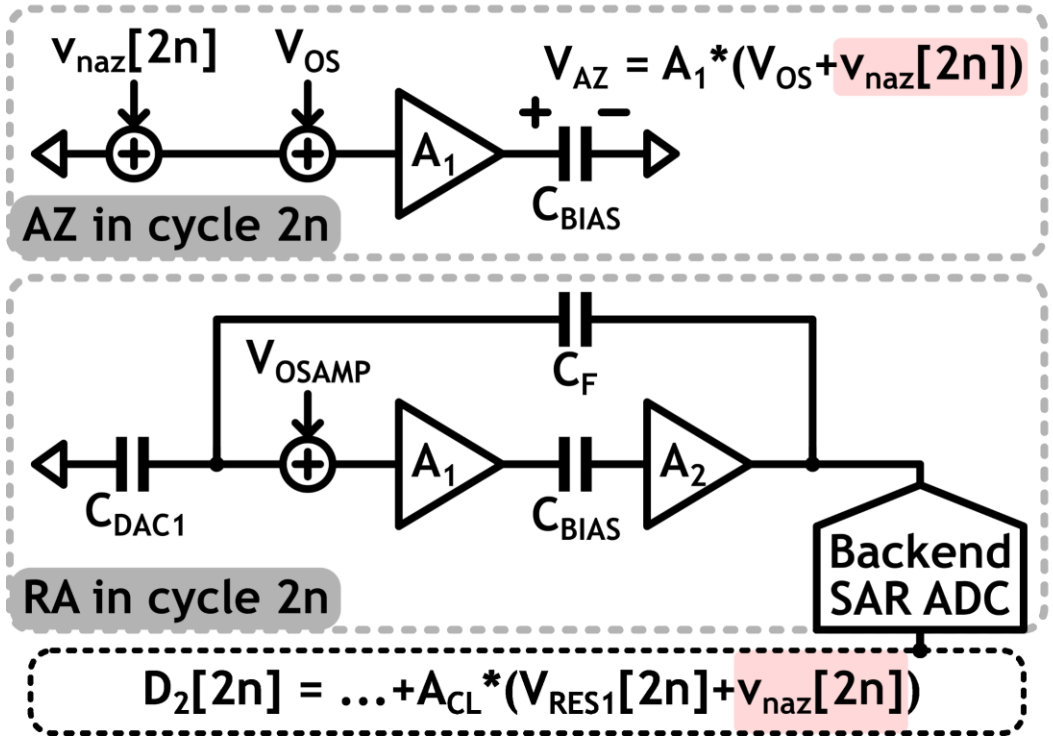




- 12b SAR conversion (Async)

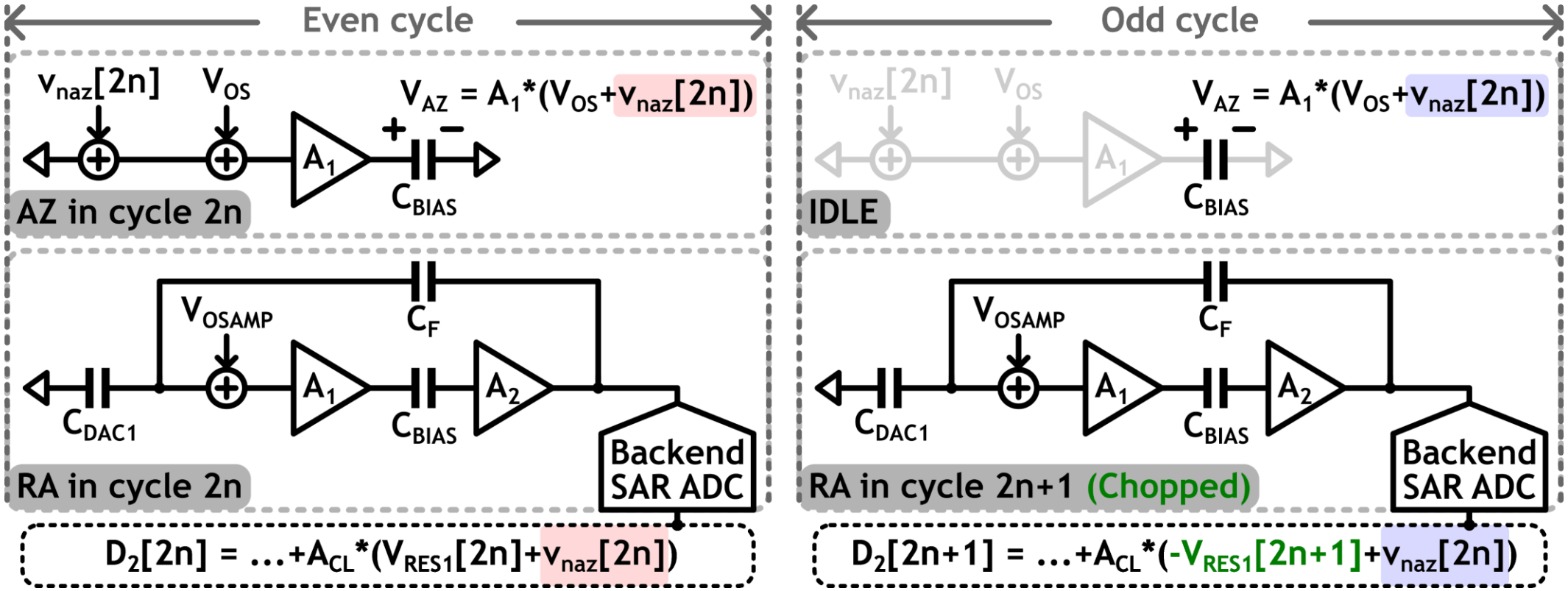
Conventional Auto-zero

- AZ Performed in Every Cycle
  - AZ-induced amplifier thermal noise ☹



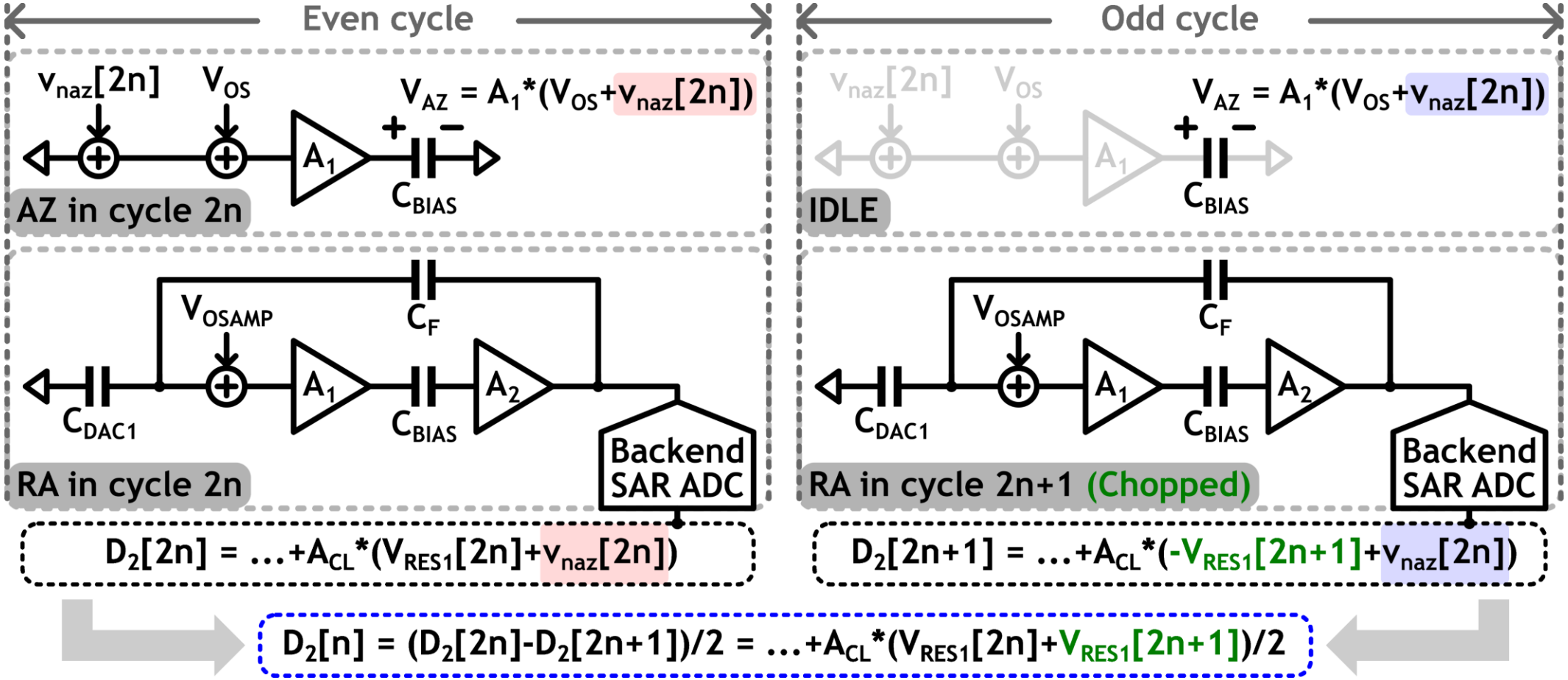
System-level Chopping-assisted Auto-zero

- AZ Skipped in the Odd Cycle with Reversed Input



System-level Chopping-assisted Auto-zero

- AZ Skipped in the Odd Cycle with Reversed Input
  - Noise eliminated by output averaging ☺

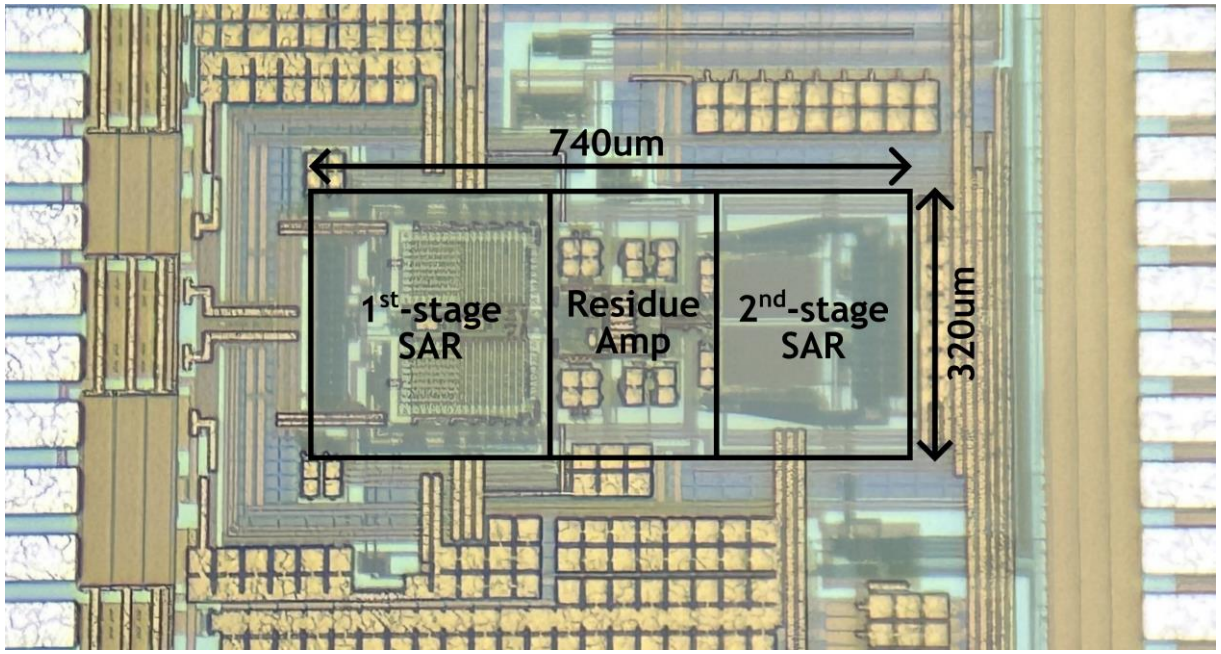


Outline

- Motivation
- Proposed Closed-loop Dynamic Amplifier
  - Floating Gm-ratio-based Amplifier
  - Charge-pump-assisted Cascoded FIA
- Pipelined SAR ADC Circuit Implementation
  - System-level Chopping-assisted Auto-zero
- Measurement Results
- Conclusion

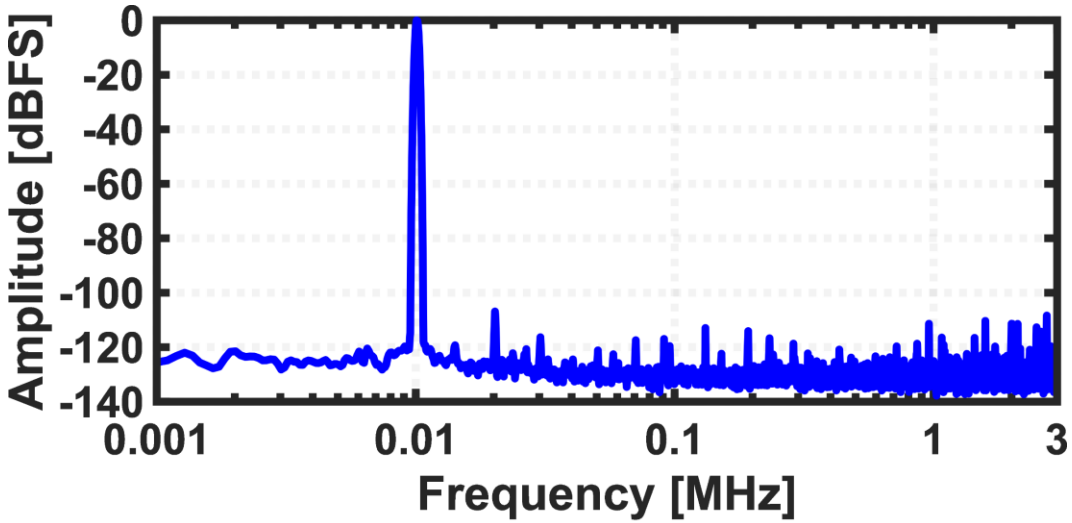
Chip Micrograph

- 180nm CMOS
- Active Area: 0.237mm^2
- Supply: 1.8V/1.2V
- Conversion Speed: 6MS/s
- One-time Foreground Calibration for Capacitor Mismatches

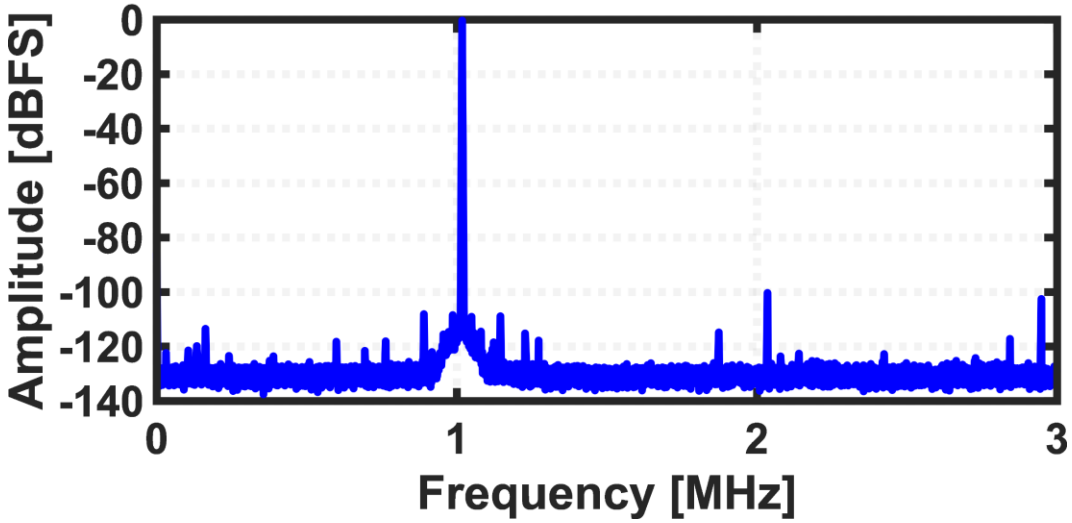


Measured Spectrum

- Low-frequency Input
  - 64k-points, 10x averaging
  - Fin = 10kHz
  - SFDR = 107.1dB
  - SNDR = 89.2dB

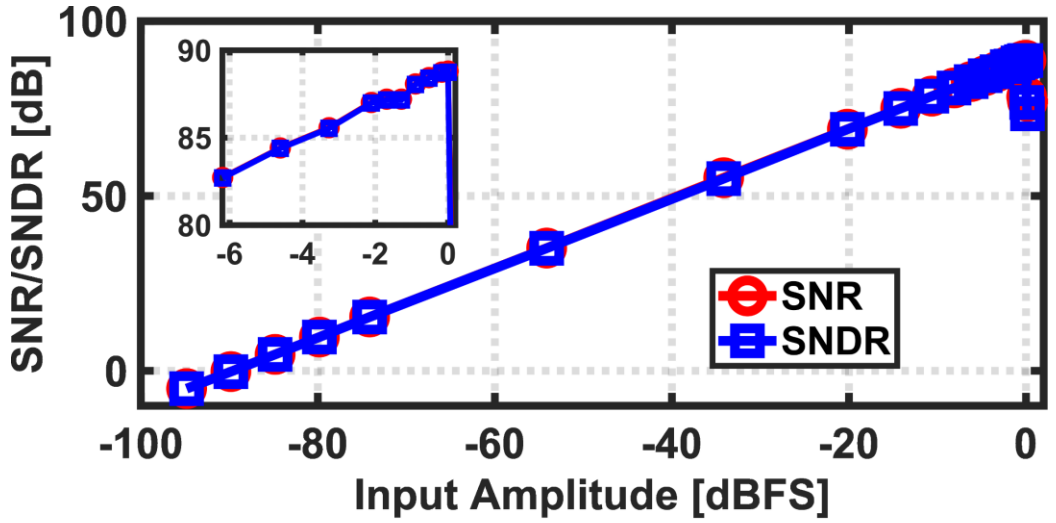


- Medium-frequency Input
  - 64k-points, 10x averaging
  - Fin = 1MHz
  - SFDR = 100.5dB
  - SNDR = 88.8dB

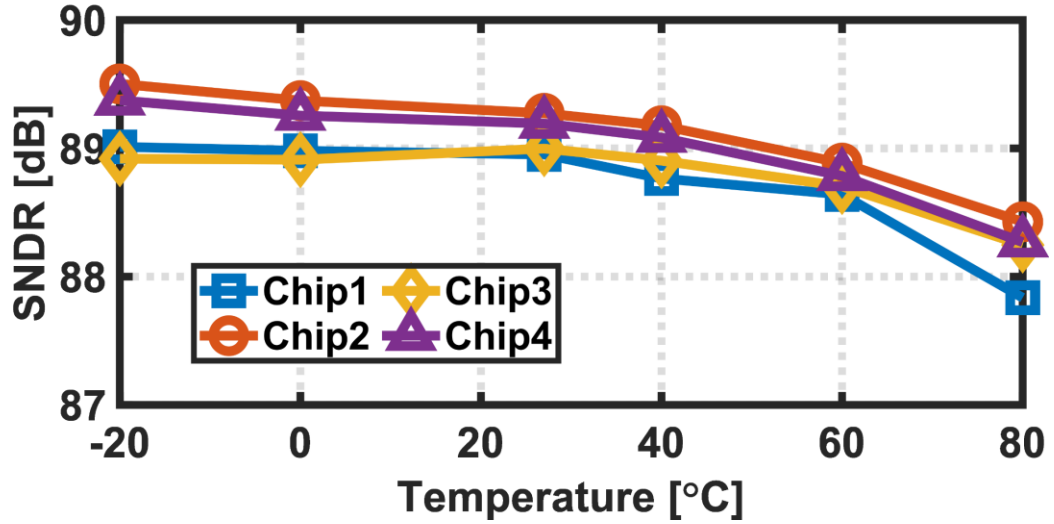


Measured DR & SNDR Variations

- Performance vs. Input Amp.
  - SNDR,SNR
  - DR = 89.6dB

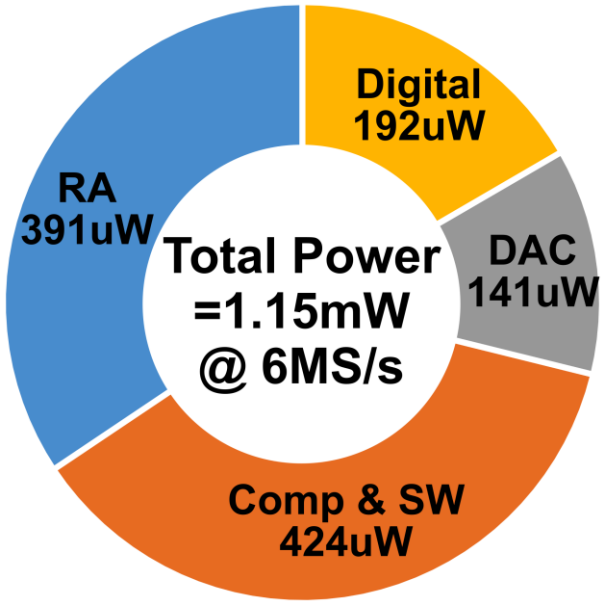


- SNDR vs. Temperature
  - 4 chips
  - -20 ~ 80 ℃
  - ΔSNDR < 2dB



Measured Power Breakdown

- Power Breakdown
  - Supply = 1.8V(Analog) / 1.2V(Digital)
  - Conversion speed = 6MS/s
  - Total power = 1.15mW



Comparison with Prior Works

	VLSI-14 Bannon	VLSI-17 Hummerston	ISSCC-24 Bodnar	ISSCC-23 Li	ISSCC-21 Tang	This work
Architecture	Pipelined SAR	Pipelined SAR	Pipelined SAR	Pipelined SAR	Pipelined SAR	**Pipelined SAR**
Residue Amplifier	Closed-loop OTA	Closed-loop OTA	Closed-loop OTA	Ring Amplifier	3-stage FIA	**FGA+CP-FIA**
Process [nm]	180	180	40	180	40	**180**
Speed [MS/s]	5	2	40	12	40	**6**
Supply [V]	5/1.8	5/1.8	3.3/1.1	3.3/1.8	1.2	**1.8/1.2**
Area [mm^2]	5.74	3.87	0.9	0.6	0.056	**0.237**
Power [mW]	30.52	11.35	50	9.43	0.82	**1.15**
SNDR(SNR*) [dB]	98.6 @1kHz	100* @100kHz	93.7* @1kHz	91.3 @4MHz	75.7 @18MHz	**88.8 @1MHz**
DR [dB]	100.2	100.5	94.2	94.1	79.5	**89.6**
FoMw [fJ/c-s]	87.7	69.4	31.6	26.2	4.1	**8.5**
FoMs [dB]	177.7	179.5	179.7	179.3	179.6	**183.0**

Conclusion

- Proposed Pipelined SAR ADC with a Closed-loop Dynamic Amplifier
  - 1st-stage FGA ➔ High speed and robust stability
  - 2nd-stage CP-FIA ➔ High linearity and slew rate
  - System-level chopping-assisted AZ ➔ low 1/f noise, no thermal noise penalty

  - Suitable for high-precision and low-latency readout (88.8dB-SNDR@1MHz) ☺
  - High energy efficiency (183dB FoMs) ☺

Thanks for your attention!